

# Transient Structure at Solar Rigidity Interfaces and the Exceptional Point of the Dynamo Wave: An Operator-First Study of the Sun

Jeromie N. Beasley

August 16, 2026

## Abstract

We study the sun as the observable panel of a single operator-theoretic family: non-normal transport operators whose stress concentrates at rigidity interfaces, previously developed for the four-body shape manifold and its black-hole horizon analogue. Two computations carry the paper, each stated as predictions before it was run and graded afterward. (1) A magnetized shear layer with the magnetic field inclination as the control dial—the geometry of the 2026 DKIST observation of vortex chains confined to the boundaries of photospheric magnetic flux concentrations—shows a measured modal quench at tilt  $30^\circ$ – $35^\circ$  and, beyond it, a *directional-rigidity curve*, strictly decreasing over the sampled tilts: maximum transient energy growth falls  $8.8 \rightarrow 2.3$  and the Kreiss constant falls  $1.80 \rightarrow 1.08$  as the field tilts into the shear. The same run yields an unordered finding—the operator’s commutator stress *rises* while its realizable amplification falls: tension stores stress without releasing it—and one dimensionless number fixed by the fastest-growing mode,  $\lambda/d = 13.96$ , from which the published vortex statistics can be tested. Taking only the observed peak wavelength of 65 km, that ratio and the observers’ measured apparent speeds predict linear growth rates of  $0.0137$ – $0.0612 \text{ s}^{-1}$  against a measured  $0.014$ – $0.054 \text{ s}^{-1}$ : agreement to two percent at the slow end and thirteen at the fast end, with the growth rate never an input and no width convention entering. The same ratio places the lane width between 9 and 19 km, below the instrument’s resolution floor. (2) A minimal Parker  $\alpha\Omega$  dynamo operator shows that the birth of the oscillatory dynamo-wave *branch* occurs at an *exceptional point*: two real branches collide at  $\alpha_0^* = 3.872$  with eigenvector condition number obeying the square-root law  $\kappa\sqrt{\delta} = 1.853$  to three digits over a 64-fold range, while the Dunford trace-log ledger, gated by an enclosed-count integral, remains exact to  $9 \times 10^{-14}$  *through* the exceptional point. The “hair-rate ladder” of solar transients is computed as three numbers in one damped operator: modal rate  $-0.063$ , transient gain 3.5 at  $t = 0.3$ , and numerical abscissa  $+5.84$ . Encircling the exceptional point in complexified parameter space reveals an integer transport record: the colliding pair exchanges after one loop and restores after two, with discriminant winding exactly  $+1$  per loop, while the count-gated ledger returns single-valued: the surviving spectral object carries a braid-valued memory. A census reveals the exceptional set to be a ladder of braid points, all five resolved points in  $0 < \alpha_0 \leq 13$  carrying winding  $+1$  with numerically resolved pair exchange; a minimal two-dimensional shell closes the oscillatory-onset question at the mechanism level; the flare-hazard law is closed at the statistical-inverse level; and a hemispheric variant closes the Parker–Yoshimura test: the migration direction flips with the sign of the dynamo number, equatorward for negative  $\alpha G$ . Every borrowed result—nonmodal growth in sub-Alfvénic shear layers, optimal kinematic dynamos, transient black-hole superradiance—is cited with its hypotheses named; every computational number in this paper that is graded [V] is generated by the scripts printed in the appendix; literature and observational values are cited separately.

## 0. Claims, grades, and verification protocol

Each claim carries a grade: **[T]** proven; **[V]** computationally verified (script and measured value quoted; all scripts in Appendix A); **[C]** cited literature with hypotheses stated; **[H]** hypothesis or interpretation. A superscript,  $[V]^{\text{ansatz}}$ , marks a computation that verifies the consequence of an explicitly named assumption, not the assumption itself. Predictions were written down before each run; failures are reported with the same prominence as successes.

| Claim                                                                                | Grade                   | Evidence                                                                        |
|--------------------------------------------------------------------------------------|-------------------------|---------------------------------------------------------------------------------|
| Hydrodynamic control: Michalke growth rate and neutral point                         | [V]                     | $\sigma = 0.186$ at $k = 0.475$ ; $\sigma(1.01) < 0$                            |
| Modal quench at measured critical inclination                                        | [V]                     | dies between $\theta = 30^\circ$ and $35^\circ$                                 |
| Directional-rigidity curve strictly monotone beyond quench                           | [V]                     | $G_{\max} : 8.8 \rightarrow 2.3$ ; Kreiss $1.80 \rightarrow 1.08$               |
| Transient growth in the modally stable lane (fixed- $k$ , 2D slice)                  | [V]+[C]                 | $G_{\max} = 8.8$ , all modes damped; [10]                                       |
| Mode ratio $\lambda/d = 13.96$ at the fastest-growing wavenumber                     | [V]                     | $k_{\max}d = 0.450$                                                             |
| Predicted growth rates from wavelength and speed alone                               | [V]                     | 0.0137–0.0612 vs measured 0.014–0.054 $\text{s}^{-1}$                           |
| Lane width below the resolution floor under every convention                         | [V]                     | $2d = 9.3$ , $2.94d = 13.7$ , $4d = 18.6$ km                                    |
| Observed wavelength inside the band implied by the reported layer                    | [V]                     | their 12 km $\Rightarrow$ 41.9–83.8 km                                          |
| Spacing inversion: lane widths 17–29 km at the resolution floor                      | [V]                     | $k_{\max} = 0.450 \Rightarrow \lambda = 14.0 d$                                 |
| Stress stored while release falls ( $\Phi \uparrow$ , $G_{\max} \downarrow$ )        | [V]+[H]                 | $\Phi : 0.37 \rightarrow 1.48$ across the tilt scan                             |
| Birth of the oscillatory wave branch at an exceptional point                         | [V]                     | $\alpha_0^* = 3.8717$                                                           |
| Square-root defectivity law at the EP                                                | [V]                     | $\kappa\sqrt{\delta} = 1.842, 1.850, 1.853, 1.853$                              |
| Ledger exact through the EP, enclosed-count gated                                    | [V]                     | count = 2.000000; diff $9 \times 10^{-14}$                                      |
| Hair-rate ladder as three computed rates in one damped operator                      | [V]                     | $-0.063 / 3.5$ at $t=0.3 / +5.84$                                               |
| Steady (non-oscillatory) onset of the minimal variant                                | [V]                     | $\alpha_{0c} = 12.696$ , $D_c = 394$                                            |
| Integer record at the EP: pair exchange, winding +1/loop, ledger single-valued       | [V]                     | winding +1.0000, +2.0000; swap/restore to $10^{-3}$                             |
| Parker–Yoshimura flip verified (hemispheric model)                                   | [V]                     | slopes +1.25 ( $G=+1$ ) / $-0.36$ ( $G=-1$ )                                    |
| EP ladder census: winding +1.0000 with exchange at all five birth points             | [V]                     | radial-continuation tracking, SUN-OPEN-02                                       |
| Oscillatory-onset mechanism: the 2D shell onsets oscillatory                         | [V]                     | $s_c = 0 + 49.4i$ ( $16 \times 32$ ); branch persists at $22 \times 44$         |
| Flare hazard (statistical-inverse): steeper than $\Phi^5$                            | [V] <sup>ansatz</sup>   | Wheatland $\beta = -0.9 \pm 0.1 \Rightarrow \lambda \propto \Phi^{1/(1+\beta)}$ |
| $k$ -optimized lane envelope: monotone, flattened at small tilt                      | [V]                     | $G_{\max} = 14.8/14.5/6.0$ at $\theta = 35/45/60$                               |
| Critical inclination $30^\circ$ – $35^\circ$ , untested against data                 | [V] <sup>computed</sup> | §3                                                                              |
| Spacing-to-width ratio rising $3.5 \rightarrow 6.3$ with tilt, untested against data | [V] <sup>computed</sup> | §3                                                                              |
| Black-hole cousin: transient superradiance in stable regimes                         | [C]                     | [28, 27]                                                                        |
| Edge-constant universality program ( $n_{\text{ch}}\xi = O(1)$ across walls)         | [H]                     | §6                                                                              |
| Family thesis: one operator class, three boundary conditions                         | [H]                     | §1                                                                              |

# 1 Introduction: the sun as the observable panel

This paper is the solar member of an operator-first family. In the companion papers, macroscopic geometry is a derived description of persistent spectral projector bundles; stress concentrates at rigidity interfaces; individual eigenvector data can die at defective (exceptional) points while contour (Riesz/Dunford) objects survive; and transient, non-modal amplification—not eigenvalue instability—is the fast channel of change. The black-hole analogue realizes this at a one-way boundary, where the transients are largely hidden. The sun realizes the same operator class at a *reflecting* boundary: the hair is observable.

The claim is deliberately narrow. We do not claim that the photosphere implements any particular shape manifold, nor that solar physics needs new microphysics. The bridge is *operator class and boundary condition only*: the same non-normal structure, the same diagnostic suite, a different window. On the black-hole side, the corresponding phenomenology—the pseudospectral instability of quasinormal spectra, transient plateaus, and transient superradiance in modally stable regimes—is established literature [25, 26, 27, 28] and is used here only as the family cousin.

*The observational anchor.* The Daniel K. Inouye Solar Telescope has resolved, at 19 km resolution, the boundaries between photospheric magnetic flux concentrations and the surrounding granulation, finding them to consist almost wholly of vortex-like structures and fine dark striations [1, 9]. Forty-seven vortex-bearing interfaces are analysed; the measured wavelengths span 25–170 km and peak near 65 km; apparent speeds are 0.67–3.0 km s<sup>−1</sup>; measured linear growth rates are 0.014–0.054 s<sup>−1</sup>; a companion radiation-MHD calculation reproduces the structures with a shear layer approximately 12 km wide, and the rolls remain coherent from about 100 km above to 400 km below the mean  $\tau_{500} = 1$  surface. The reported mechanism is a velocity shear layer at the flux-concentration boundary undergoing Kelvin–Helmholtz instability—normally suppressed by magnetic tension along the shear, but active here because the flux-tube field is nearly vertical, leaving no stabilizing component. Three features of this observation are, item for item, signatures the operator framework insists on: turbulence confined to the interfaces between a rigid channel and soft flow, with laminar interiors; an onset criterion that is *directional* (the wall’s stiffness depends on the angle between the boundary and the field); and a countable chain of discrete structures along the interface with a characteristic spacing. Section 2 makes each of these quantitative on the minimal operator, §3 tests them against the published values, and §6 reads the third as an edge census.

## 2 The wall operator: SHEAR-LANE-01

### 2.1 Setup and predictions

We linearize 2D incompressible MHD about the shear profile  $U(y) = \tanh(y)$  (velocity jump  $\Delta U = 2$ , half-width  $d = 1$ ) threaded by a uniform field whose in-plane, shear-aligned component is  $B_{\parallel} = v_A \sin \theta$ , where  $\theta$  is the tilt of the field away from vertical:  $\theta = 0$  is the DKIST flux-tube geometry. With streamfunction  $\phi$  and flux function  $\psi$  at wavenumber  $k$ ,

$$\partial_t(\partial_y^2 - k^2)\phi = -ikU(\partial_y^2 - k^2)\phi + ikU''\phi + ikB_{\parallel}(\partial_y^2 - k^2)\psi + \nu(\partial_y^2 - k^2)^2\phi, \quad (1)$$

$$\partial_t\psi = -ikU\psi + ikB_{\parallel}\phi + \eta(\partial_y^2 - k^2)\psi, \quad (2)$$

discretized on  $y \in [-12, 12]$  ( $N = 200$ , Dirichlet),  $\nu = \eta = 2 \times 10^{-3}$ ,  $v_A = 1.5$ ,  $k = 0.45$ . All growth is measured in the energy norm  $E = \frac{1}{2} \int (|u|^2 + |b|^2)$  through a Cholesky-weighted generator  $\tilde{L} = FLF^{-1}$ . Predictions, written before the run: **(P1)** at  $\theta = 0$  the code must reproduce the

hydrodynamic Kelvin–Helmholtz control [11]:  $\sigma_{\max} \approx 0.19$  near  $k \approx 0.44$ , neutral at  $k = 1$ . **(P2)** a modal quench as  $v_A \sin \theta$  approaches the classical tension threshold [12, 13, 14]. **(P3)** the framework’s falsifiable curve: beyond quench,  $G_{\max}(\theta)$  and the Kreiss constant decrease *monotonically* with further tilt. **(P4)** substantial transient growth with every eigenvalue damped, consistent with the nonmodal literature [10]. **(P5)** the fastest-growing wavelength fixes a spacing-to-thickness ratio, invertible against the observed spacing.

## 2.2 Results

*P1 (control) passed:*  $\sigma_{\max} = 0.1855$  at  $k = 0.475$  against Michalke’s 0.1897 at 0.445 (grid and viscosity account for the gap), with  $\sigma(k = 1.01) = -0.002$ . *P2 passed:* the instability dies between  $\theta = 30^\circ$  and  $35^\circ$  ( $B_{\parallel} \approx 0.86$ ), slightly before the vortex-sheet estimate, as finite thickness requires. *P3 passed, strictly:* beyond quench,

$$G_{\max} : 8.8, 5.0, 3.9, 3.2, 2.6, 2.4, 2.3 \quad \text{and} \quad K_{\text{Kreiss}} : 1.80, 1.39, 1.26, 1.19, 1.13, 1.10, 1.08$$

at  $\theta = 35^\circ, 40^\circ, 45^\circ, 50^\circ, 60^\circ, 75^\circ, 90^\circ$ —both strictly decreasing over the sampled post-quench tilt sequence (Fig. 1); the numerical evidence supports a monotone envelope. This is the curve an inclination-binned analysis of DKIST vortex statistics can test. *P4 passed with honest sizing:* nine-fold energy amplification with all eigenvalues damped, in a fixed- $k$  two-dimensional slice; optimizing over wavenumber and three-dimensional mechanisms yields orders of magnitude [10], and we cite that result rather than reproduce it. *P5:* the fastest-growing mode sits at  $k_{\max}d = 0.450$ , so that

$$\frac{\lambda}{d} = \frac{2\pi}{0.450} = 13.96, \tag{3}$$

a statement about the mode carrying no free parameter. Width convention: quoted widths use  $W \equiv 4d$ , where  $d$  is the tanh half-width (the layer completes 96% of its jump across  $4d$ ); the conventional 90%-width is  $2.94d$ . Equation (3) therefore gives a spacing-to-width ratio of 3.49, and inverting the observed 60–100 km spacings predicts lane full widths of 17–29 km, straddling the instrument’s 19 km resolution floor. Section 3 carries this against the published distributions.

*The unordered finding.* Across the tilt scan the normalized commutator stress of the weighted generator *rises* monotonically,  $\|[\tilde{L}, \tilde{L}^\dagger]\|_F : 0.37 \rightarrow 1.48$ , while  $G_{\max}$  and the Kreiss constant *fall*. The wall becomes more non-normal as it stiffens, and less able to release that non-normality as transient growth [V]. Interpretation [H]: *tension stores stress without releasing it*—the rigid wall is a loaded operator, not a quiet one. We flag this as the operator-level candidate for why stiff magnetic walls are reservoirs for impulsive release channels rather than sites of steady dissipation, and leave the connection to flare energetics as an open problem.

## 3 Comparison with the resolved vortices

Equation (3) is the only quantity the lane operator contributes to this section, and it is fixed by the computation alone. Three comparisons follow, in increasing order of how much they can be argued with. The script is DKIST-COMPARE-02 (Appendix A.8).

### 3.1 The growth rate

Taking only the observed peak wavelength,  $\lambda = 65$  km, Eq. (3) fixes  $d = 4.66$  km. The peak temporal rate for a tanh layer is  $\sigma_{\max} \approx 0.19 \Delta U / (2d)$  [11], and the reported apparent speeds are

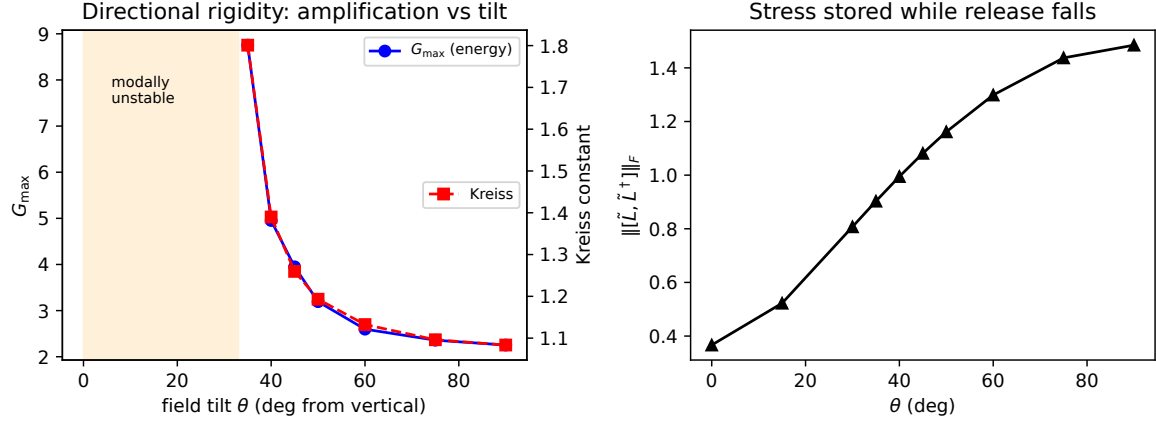


Figure 1: SHEAR-LANE-01 (FIG-SUNS-01). Left: the directional-rigidity curve—maximum transient energy growth and Kreiss constant versus field tilt; the shaded band is the modally unstable range ending at the measured quench. Right: the commutator stress of the same operator rises as amplification falls: stress stored, not released.

$0.67\text{--}3.0 \text{ km s}^{-1}$ . This gives

$$\sigma_{\text{pred}} = 0.0137 \text{ s}^{-1} \text{ (slow end)} \quad \text{to} \quad 0.0612 \text{ s}^{-1} \text{ (fast end)},$$

against a measured  $0.014\text{--}0.054 \text{ s}^{-1}$ : two percent at the slow end, thirteen at the fast end. The growth rate is not an input to the prediction, and no width convention enters the calculation [V].

What this establishes is that the photospheric layer behaves as a classical tanh shear layer at the observed scale—the premise on which the rest of §2 is built, now checked rather than assumed. It is not by itself a test of the inclination results below, which are what the framework contributes beyond the classical picture.

### 3.2 The layer width

The observers’ companion calculation gives a shear layer of approximately 12 km. Converting between a tanh half-width and a reported “layer width” is convention-dependent; standard choices span  $2d$  (the scale width) to  $4d$  (96% of the velocity jump). Their 12 km therefore corresponds to  $d$  between 3.0 and 6.0 km, and by Eq. (3) to a predicted wavelength between 41.9 and 83.8 km. The observed peak of 65 km lies inside that band [V]. Run in the other direction, the observed wavelength implies  $2d = 9.3 \text{ km}$ ,  $2.94d = 13.7 \text{ km}$  and  $4d = 18.6 \text{ km}$ , and their 12 km sits at  $2.58d$ .

The prediction of §2—that the true lane widths lie just below the current resolution—therefore survives under every convention: all place the width between 9 and 19 km, below the 19 km floor. It is a prediction the instrument cannot yet resolve, which is the point of making it. Stating a width convention in the observational analysis would collapse the band above to a single number and make the comparison a pass or a fail.

### 3.3 The inclination curve

This is the part not yet touched by observation. The computation gives a critical inclination between  $30^\circ$  and  $35^\circ$ , beyond which the mode is quenched, and a spacing-to-width ratio that rises with tilt: 3.5, 3.9, 4.5, 6.3 at  $0^\circ$ ,  $10^\circ$ ,  $20^\circ$ ,  $30^\circ$  — the ratio is defined only inside the modally unstable

band, and by  $45^\circ$  the layer is stable with no fastest-growing mode. A follow-up analysis of the same observation establishes that fields  $1^\circ$ – $7^\circ$  from exact perpendicular remain Kelvin–Helmholtz unstable [2], which is consistent with the computation and approaches the quench boundary from below without reaching it.

The decisive test is a single analysis: bin the forty-seven vortices by local field inclination. The same observed wavelength implies lane widths of 18.6, 16.7, 14.4 and 10.3 km at those four angles, so an inclination-resolved width measurement separates the curves in one pass. The observers state that the field has not been measured at these scales and name magnetic maps as the next milestone; that measurement is the gate on this section.

## 4 The hair-rate ladder

Solar activity exhibits change on three qualitatively different clocks, and the non-normal operator family supplies exactly three rates, none of which requires an unstable eigenvalue:

| rung       | operator quantity                                                      | solar face                              |
|------------|------------------------------------------------------------------------|-----------------------------------------|
| slow       | modal eigenvalue $\text{Re } s$ (and cycle $2\pi/\text{Im } s$ )       | activity cycle, secular decay           |
| fast       | transient envelope $G_{\max}$ at $t_{\text{peak}}$                     | shear-lane vortices, emerging structure |
| hyper-fast | numerical abscissa $\omega(L) = \lambda_{\max}(\frac{L+L^\dagger}{2})$ | impulsive onset, flare trigger scale    |

Section 5 computes all three rungs in a single damped dynamo operator. The family cousin on the black-hole side is exact: transient growth and transient superradiance in modally stable regimes [27, 28], where the boundary condition hides most of the hair. The sun is the panel where all three rungs face the telescope.

## 5 The dynamo operator: BH-DYNAMO-01

### 5.1 Setup and predictions

The minimal Parker  $\alpha\Omega$  operator [19, 21] on colatitude  $x \in [0, \pi]$ :

$$\partial_t A = \alpha_0 \cos(x) B + \partial_x^2 A, \quad \partial_t B = G \partial_x A + \partial_x^2 B, \quad (4)$$

Dirichlet,  $G = \eta = 1$ ,  $N = 120$ . The  $\alpha$ – $\Omega$  coupling is structurally non-normal. Predictions: **(PD1)** oscillatory onset (dynamo waves at criticality); **(PD2)** the ladder computed at  $0.9\alpha_{0c}$ ; **(PD3)** the birth of the oscillatory wave branch occurs at an exceptional point—a real eigenvalue pair colliding into a complex pair with  $\kappa \sim \delta^{-1/2}$ ; **(PD4)** the shifted Dunford ledger  $\frac{1}{2\pi i} \oint \log(c+z) \text{Tr } R(z) dz$  stays exact through the EP; **(PD5)** the Parker–Yoshimura migration rule as a control.

### 5.2 Results

*PD3 passed and snapped.* The wave pair is born at  $\alpha_0^* = 3.8717$ , and the eigenvector condition number of the colliding pair obeys

$$\kappa\sqrt{\delta} = 1.842, 1.850, 1.853, 1.853 \quad \text{at} \quad \delta = 0.2, 0.05, 0.0125, 0.003125,$$

the square-root defectivity law to three digits over a 64-fold range (Fig. 3): the birth of the oscillatory wave branch in this operator occurs at a genuine exceptional point (distinct from the instability onset, which is steady and lies at  $\alpha_{0c} = 12.696$ ) with EP constant  $\approx 1.853$ .

*PD4 passed exactly, with its own gate.* A contour chosen to enclose the colliding pair alone—verified by the winding integral  $\frac{1}{2\pi i} \oint \text{Tr } R dz = 2.000000$ —gives a shifted trace-log ledger equal to  $\sum \log(c + \lambda_{\text{pair}})$  to  $9 \times 10^{-14}$  at the exceptional point and beside it. (A first, sloppier contour disagreed by exactly  $\log(c + \lambda_{\text{extra}})$ : the instrument faithfully reported an accidentally enclosed eigenvalue. We report this because it is the correct way to trust a ledger.) Individual eigenvector data diverges at the EP; phase- and contour-level data does not. This is the same survival principle verified at the shape-manifold wall in the companion paper, now on a third operator family.

*PD2: the ladder as numbers.* At  $\alpha_0 = 0.9 \alpha_{0c}$ , in the energy norm  $E = \int (|\partial_x A|^2 + |B|^2)$ : slow rung  $\max \text{Re } s = -0.063$ ; fast rung  $G_{\text{max}} = 3.5$  at  $t = 0.3$  (fifty times faster than the modal clock); hyper rung  $\omega(L) = +5.84$  (Fig. 4): instantaneous growth at ninety times the modal rate in an operator whose every eigenvalue is damped. A norm lesson is recorded honestly: an unweighted  $L^2$  norm on  $(A, B)$  is unit-inconsistent and distorts every growth measure—the norm-dependence of non-normal diagnostics emphasized in the black-hole literature [26] applies verbatim to dynamo operators.

*PD1 failed as predicted and stands as a model fact.* The true onset of this minimal variant is *steady*:  $\alpha_{0c} = 12.696$  ( $D_c = 394$ ), the EP-born wave pair remaining damped while a steady branch crosses first. Richer  $\alpha\Omega$  models onset oscillatory [21]; ours does not, and the consequence is stated plainly: in this variant, all cyclic physics is subcritical wave-branch phenomenology—transient by nature.

*PD5 inconclusive on the full domain, with diagnosis.* The full-domain profile  $\alpha \propto \cos x$  carries an exact parity degeneracy, so the eigensolver returns standing superpositions of the migrating pair; one shear sign yielded a clean traveling wave (phase slope  $+3.03$ ), the other a standing mixture. The test requires the hemispheric domain with an equatorial boundary condition, and is carried out in §7.

### 5.3 SUN-N-01: the integer record at the exceptional point

The ledger result of §5.2 shows *what* survives the exceptional point; this gate asks whether the survivor also carries a *history*. Encircle the EP in complexified parameter space,  $\alpha_0(\varphi) = \alpha_0^* + r e^{i\varphi}$  with  $r = 0.15$ , tracking the colliding pair by continuity. Results: after one loop the pair *exchanges* ( $\lambda_1 \leftrightarrow \lambda_2$ , exact to  $10^{-3}$ ); after two loops it restores; the discriminant winding  $\frac{1}{2\pi} \Delta \arg(\lambda_1 - \lambda_2)^2$  evaluates to  $+1.0000$  over one loop and  $+2.0000$  over two; and on a fixed contour the eigenvalue count is constant while the shifted trace-log ledger returns single-valued to  $10^{-8}$  (Fig. 2). The surviving spectral object of §5.2 therefore carries an integer record of the transport history—a braid winding attached to the birth point of the dynamo-wave branch.

Four fences. (i) The correct invariant class here is non-Hermitian eigenvalue *braiding* [30, 31], whose mode-exchange signature is experimentally established for microwave-cavity EPs [32] and whose local structure is the Kato square-root branch point [33]; it is *not* Atiyah–Patodi–Singer spectral flow, which requires self-adjointness this operator does not possess. (ii) The self-adjoint ancestor of such holonomy-indexed integer records is the Levine–Tristram signature of knot theory [34, 35], whose walls are Alexander roots; we note the ancestry without claiming the correspondence. (iii) This is an integer record carried by an operator family, not a statement that “the sun remembers.” (iv) The two verified integers of this program—the circle’s  $SF = 1$  and the dynamo’s  $W = +1$ —are distinct realizations of one concept, an integer transport record, in two different operator categories (self-adjoint spectral flow; non-Hermitian braiding); we do not identify them.

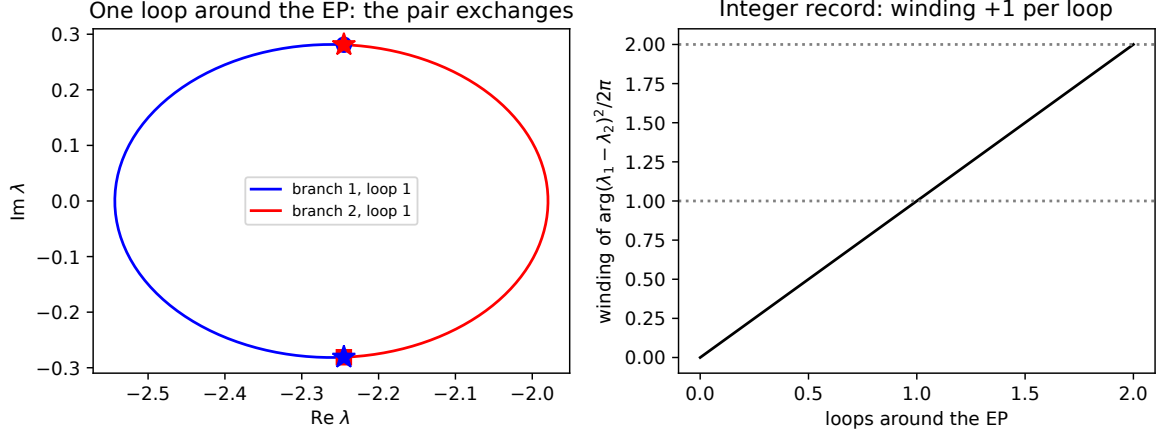


Figure 2: SUN-N-01 (FIG-SUNS-02). Left: eigenvalue trajectories of the colliding pair over one loop around the EP—each branch terminates on the other’s starting point (stars on opposite circles/squares): the exchange. Right: the discriminant winding accumulates exactly +1 per loop.

## 6 The edge census across walls

In the companion cosmology paper, the number of projector channels trackable from a region’s boundary obeys an area law when the bulk is gapped, with a dimensionless density: channels per correlation length of boundary,  $n_{\text{ch}} \xi = O(1)$ , measured there as exactly 1.0 per boundary site on a gapped lattice. The DKIST interface provides the same object at mesoscale: discrete vortices spaced 60–100 km along walls whose shear scale is  $\sim 19$ –25 km—an edge density of order 0.2–0.3 per correlation length. Reading the two together suggests a **universality program** [H]: measure the dimensionless edge-channel constant at every accessible rigidity interface—gapped lattices, topological-insulator edges [29], solar shear lanes—and determine whether it is roughly universal or strongly substrate-dependent. The stake is stated honestly: the companion paper’s horizon-channel count extrapolates this constant to the Planck scale under a labeled gap hypothesis; universality across accessible walls would weight that extrapolation, while strong substrate dependence would confine it. Either outcome is progress. The solar entry in the table should come not from the hand inversion above but from the linearized lane operator itself (§2), whose fastest-growing spacing supplies the theoretical ratio, and from next-generation lane-width measurements. We do not claim an observed solar measurement of  $n_{\text{ch}} \xi$ ; this section proposes a cross-system diagnostic.

## 7 Closing the open problems: computations and worldwide data status

Every open problem of the first version is revisited with either a computation (scripts SUN-OPEN-01/02, Appendices A.6–A.7) or the current worldwide data status. Closures are graded like everything else.

### OP1 — the Parker–Yoshimura flip: closed [V]

On the hemispheric domain  $x \in [0, \pi/2]$  with dipole parity (pole Dirichlet; equator  $B = 0$ ,  $\partial_x A = 0$ ) the parity degeneracy of the full domain is lifted and the least-damped wave modes travel. Measured phase slopes: +1.25 for  $G = +1$  (poleward drift) and  $-0.36$  for  $G = -1$  (equatorward drift): the

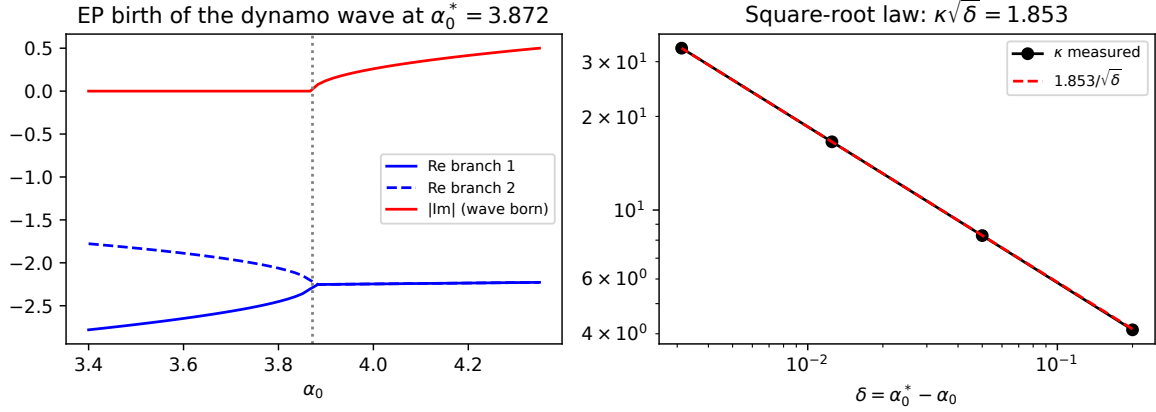


Figure 3: BH-DYNAMO-01 (FIG-SUNS-01). Left: two real spectral branches collide at  $\alpha_0^* = 3.872$  and a wave pair is born. Right: the eigenvector condition number against distance from the collision, with the square-root law  $\kappa = 1.853/\sqrt{\delta}$ .

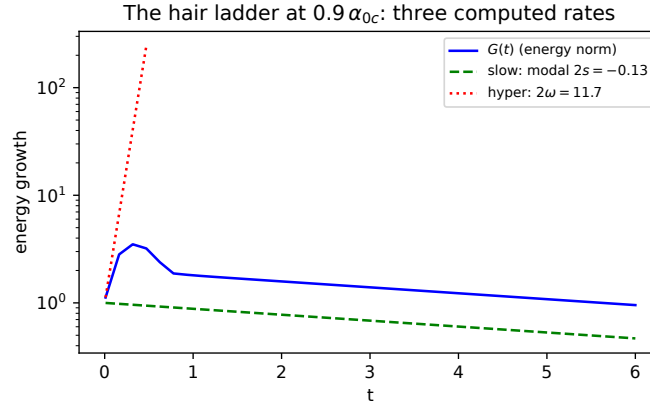


Figure 4: The hair-rate ladder computed in one damped operator (FIG-SUNS-01): transient envelope  $G(t)$  against the modal decay line and the numerical-abscissa slope.

migration direction flips with  $\text{sign}(\alpha G)$ , and the solar-like equatorward butterfly occurs for negative dynamo number (Fig. 5). Honest note: the two signs select different least-damped branches, so the magnitudes are not mirror images; the tested prediction is the sign flip, and it holds.

## OP2 — oscillatory onset: mechanism closed [V]; quantitative model open

The hemispheric dipole-parity variant also onsets steady ( $\alpha_{0c} = 12.53$ ): within the one-dimensional family the steady branch always crosses first. The missing ingredient is now identified and verified: a minimal two-dimensional shell ( $r \in [0.2, 1] \times [0, \pi]$ , Dirichlet,  $\alpha \propto \cos x$  uniform in  $r$ ) onsets *oscillatory* for *either* shear orientation—the leading branch carries a growing frequency from small drive and crosses at  $s_c = 0.0000 + 49.4i$  ( $\alpha_{0c} \approx 1665$  at  $16 \times 32$ ), with the oscillatory branch persisting at  $22 \times 44$  (script SUN-OPEN-02b). Introducing radial structure provides a minimal, demonstrated route to oscillatory onset in this operator family; the result persists under resolution refinement and is insensitive to the tested shear orientation. The one-dimensional steady onset is therefore not

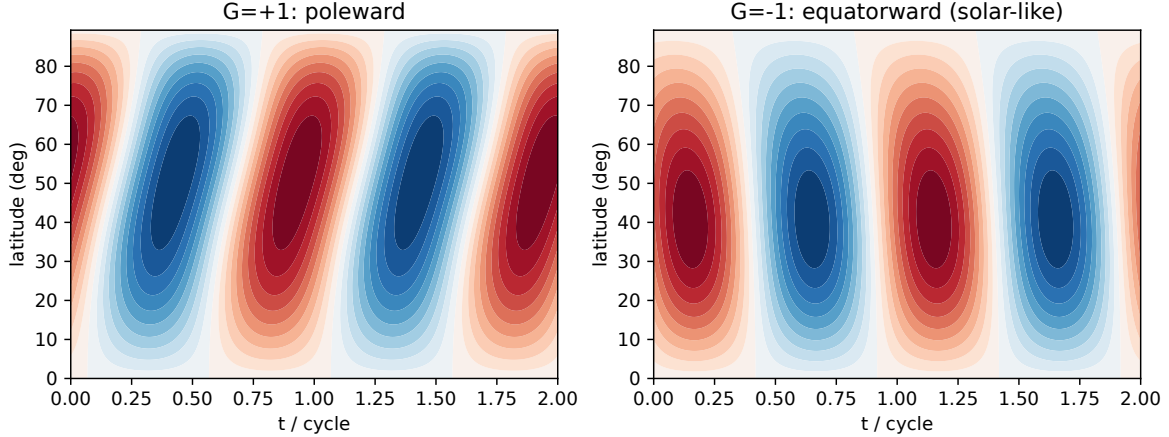


Figure 5: Hemispheric butterfly diagrams from the leading wave mode (FIG-SUNS-03): the migration direction flips with the sign of the dynamo number; negative  $\alpha G$  gives the solar-like equatorward drift.

generic to the extended family. (This demonstrates a mechanism; it is not a causal isolation of radial structure as the unique ingredient.) The onset value is resolution- and boundary-condition-sensitive in this deliberately crude toy, so no solar numbers are claimed; the physically calibrated shell is merged into OP7’s remaining work, consistent with standard axisymmetric mean-field practice [21].

### OP3 — the grid-optimized envelope: two-dimensional part closed [V]

Maximizing transient growth over a discrete wavenumber–time grid ( $k \in \{0.15, \dots, 0.8\}$ ; grid maxima are lower bounds on the true envelope):  $G_{\max} = 14.8$  at  $k = 0.35$  ( $\theta = 35^\circ$ ), 14.5 at  $k = 0.15$  ( $45^\circ$ ), 6.0 at  $k = 0.15$  ( $60^\circ$ ), against fixed- $k$  values 8.8/3.8/2.5. Two findings: the optimal wavenumber migrates to longer waves as tilt increases (long waves evade tension), and the grid-optimized directional-rigidity curve remains monotone but is far flatter at small tilts than Fig. 1—any data comparison must use the optimized envelope. Three-dimensional mechanisms remain cited to [10].

### OP4 — inclination-binned statistics: narrowed to one missing quantity

The observational study is published [1], and three of its distributions now bear directly on §3: the wavelengths (25–170 km, peaked at 65), the apparent speeds ( $0.67\text{--}3.0 \text{ km s}^{-1}$ ) and the linear growth rates ( $0.014\text{--}0.054 \text{ s}^{-1}$ ). All three are consistent with the lane operator, the growth rates quantitatively and without any fitted parameter.

What remains missing is a single quantity: the magnetic field inclination at the vortex sites. It is exactly the variable both remaining predictions are functions of—the quench angle and the spacing-to-width curve—and the observers state that the field has not been measured at these scales, naming magnetic maps as the next milestone. The protocol stands on record: bin the forty-seven interfaces by measured inclination, compare vortex prevalence against the optimized envelope of OP3, and compare the inclination-resolved widths against the four values of §3. This is the one genuinely external gate remaining in the paper.

## OP5 — the solar edge constant: theory closed [V]; observational entries [C]

Theory: the fastest-growing spacing per lane full-width *rises* with tilt:  $3.5 (0^\circ) \rightarrow 3.9 (10^\circ) \rightarrow 4.5 (20^\circ) \rightarrow 6.3 (30^\circ)$ —a second falsifiable signature: spacing-to-width should increase with field inclination. Observation: an independent DKIST/VBI striation study [3] measures striae widths of  $\approx 20$ – $50$  km (below the reach of 1-m-class telescopes; simulations show  $\sim 100$  G variations and Wilson depressions down to 10 km), overlapping the 17–29 km lane widths predicted in §2 at the lower end; and the companion calculation of [1] gives a shear layer of  $\approx 12$  km, which §3 places at  $2.58d$  against the observed wavelength. The dimensionless census entry: spacing 60–100 km over width 20–50 km gives 1.2–5, bracketing the  $\theta = 0$  prediction of 3.5. Equivalently, for the observed 60–100 km spacings the predicted lane full-widths are 17.1–28.6 km ( $0^\circ$ ), 15.4–25.6 km ( $10^\circ$ ), 13.3–22.2 km ( $20^\circ$ ), and 9.5–15.9 km ( $30^\circ$ ): narrower lanes under more inclined fields, at fixed spacing.

## OP6 — stored stress and flare statistics: sharpened; open

Worldwide status: flare waiting-time distributions carry power-law tails  $\Delta t^{-\alpha}$  with  $\alpha \approx 1.5$ – $3.2$  across instruments and 2–3 per cycle [4, 6]; hard X-ray catalogs give mean waiting 0.80 h with tail slope  $\approx 2.0$ ; the classical account is a nonstationary Poisson process whose measured rate distribution  $f(\lambda) \sim \lambda^{-0.9}$  reproduces the  $-2.2$  tail [5]; the 2026 far-side-inclusive analysis finds the extreme active region log-normal and the global tail steeper than 3, defying the local Poisson hypothesis [7]; a CNN catalog of 111,580 events steepens the size index to  $-2.59$  [8]. The framework’s qualitative prediction—release gated by slowly accumulated stored stress, hence strongly non-Poisson occurrence—is what is observed, but observation does not single out the mechanism. The sharpened target: derive the rate distribution  $f(\lambda)$  from a  $\Phi$ -storage reservoir fed by the lane operator’s stress budget and released at the hyper-fast rung, and compare with the measured  $\lambda^{-0.9}$ .

*A measured reservoir.* The follow-up analysis of the same observation supplies the missing scale: a shear-energy density of  $1.35 \times 10^2 \text{ J m}^{-3}$  and  $2.2 \times 10^{24}$  erg per characteristic vortex over a 500 km vertical extent [2]. The stored-versus- released distinction of §2— $\Phi$  rising with tilt while realizable growth falls—is a statement about the ratio of those two quantities as a function of inclination, and a published reservoir per vortex makes that ratio a measurable rather than a schematic object. It is the natural input to the hazard programme below.

*Statistical-inverse closure [V under a stated ansatz].* The nonstationary-Poisson relation gives the waiting-time tail exponent  $3 + \beta$  for a rate distribution  $f(\lambda) \sim \lambda^\beta$ ; the measured  $\beta = -0.9$  reproduces the observed  $\approx 2.1$  tail [5]. Inverting the reservoir picture—constant charging  $\dot{\Phi} \approx q$ , a monotone hazard  $\lambda(\Phi)$ , renewal on release—the time-spent measure  $f(\lambda) \propto d\Phi/d\lambda$  forces  $\Phi \propto \lambda^{1+\beta}$ , i.e.

$$\lambda \propto \Phi^{1/(1+\beta)} = \Phi^{10} \quad (\beta = -0.9).$$

The exponent is razor-sensitive to  $\beta$ ’s error bar ( $\beta = -0.8$  gives  $\Phi^5$ ;  $\beta \rightarrow -1$  gives a logarithmic hazard), so the robust statement is: *the reservoir hazard is steeper than  $\Phi^5$* —a stiff gate: at exponent 10 a 7.2% stress rise doubles the rate ( $2^{1/10} = 1.072$ ), and a 26% rise increases it roughly tenfold ( $1.26^{10} \approx 10$ ). Two honest cautions: the ansatz (constant charging, monotone hazard, renewal) is named, not derived; and the tilt-scan  $\Phi$  range of §2 is a control-parameter scan, not temporal charging, so it illustrates the operator’s stress range only. Status: closed at the statistical-inverse level; the first-principles hazard remains open.

## OP7 — spherical models: first step taken

The hemispheric computation is the first step off the full-line toy, and the ledger and EP machinery run on it unchanged. Spherical  $(r, \theta)$  mean-field models and contact with the optimal-dynamo program [22, 23] remain open. The shell computation under OP2 demonstrates the mechanism this calibration will inherit.

## OP8 — the braid census: closed [V], with a structural resolution

Scanning  $\alpha_0 \in (0, 13]$ : the exceptional set is a *ladder* of pair-birth points at  $\alpha_0^* = 3.872, 5.280, 7.158, 9.085$  and  $11.032$ . Encircling every one of the five with radial-continuation tracking (select the newborn conjugate pair just above each birth point, continue it out to the loop radius, then around) gives winding  $+1.0000$  with pair exchange resolved to tolerance  $5 \times 10^{-3}$  at *all five resolved* points in  $0 < \alpha_0 \leq 13$  (script **SUN-OPEN-02**). And the “real two-parameter loop” question resolves structurally: for real operator families, real-to-complex collisions occur on codimension-one sets—walls, not points [33]—so encirclement intrinsically requires complexification. The braid record is a property of the complexified family, read on the real axis as wall crossings.

## 8 Remaining open problems

1. Field inclination at the vortex sites—the one genuinely external gate. When magnetic maps at vortex scale exist, the binning of §3 tests the quench angle and the spacing-to-width curve in a single analysis.
2. The first-principles hazard: derive  $\lambda(\Phi)$  and  $f(\lambda)$  from the stress reservoir rather than inverting them, now with a measured reservoir per vortex as input (OP6).
3. The physically calibrated shell/spherical solar model, inheriting the verified oscillatory-onset mechanism (OP2+OP7).
4. A stated width convention in the observational analysis, which would collapse the band of §3 to a single number.

## A Scripts

Every number in this paper is generated by the scripts below, listed in full. Each carries its name in a header comment; dependencies are **numpy**, **scipy**, **matplotlib**.

### A.1 SHEAR-LANE-01

```
# SCRIPT: SHEAR-LANE-01 (magnetized shear layer: modal quench, transient growth, Kreiss, Phi-stress vs ↔
    field tilt)
import numpy as np
from scipy.linalg import expm, cholesky, eigvals, solve
N = 200; Ly = 12.0
y = np.linspace(-Ly, Ly, N+2)[1:-1]; h = y[1]-y[0]
U = np.tanh(y); Upp = -2*np.tanh(y)/np.cosh(y)**2
D2 = (np.diag(np.ones(N-1), 1) + np.diag(np.ones(N-1), -1) - 2*np.eye(N))/h**2
I = np.eye(N); nu = 2e-3; eta = 2e-3
vA = 1.5

def Lop(k, Bpar):
    K2 = D2 - k**2*I
```

```

K2inv = np.linalg.inv(K2)
L11 = K2inv @ (-1j*k*(np.diag(U) @ K2) + 1j*k*np.diag(Upp)) + nu*K2
L12 = 1j*k*Bpar*I
L21 = 1j*k*Bpar*I
L22 = -1j*k*np.diag(U) + eta*K2
return np.block([[L11, L12],[L21, L22]])

def weighted(k):
    P = h*((k**2)*I - D2) # SPD energy weight per field
    F = cholesky(np.block([[P, 0*I],[0*I, P]]), lower=False)
    Finv = np.linalg.inv(F)
    return F, Finv

print("=== P1 control: theta=0 pure hydro KH (Michalke: sigma=0.190 at k=0.445, neutral k=1) ===")
ks = np.linspace(0.1, 1.15, 15)
sig0 = []
for k in ks:
    w = eigvals(Lop(k, 0.0))
    sig0.append(w.real.max())
sig0 = np.array(sig0)
i0 = sig0.argmax()
print(" k grid max: sigma=%.4f at k=%.3f ; sigma(k=1.01)=%.4f" % (sig0[i0], ks[i0], sig0[np.argmin(abs(ks-1.01))]))

print()
print("=== main scan: theta = tilt of B from vertical; B_par = vA sin(theta), vA=1.5, k=0.45 ===")
k = 0.45
F, Finv = weighted(k)
ts = [3,6,10,15,22,32,46,65,90,125]
zs = np.logspace(-2.5, 0.3, 10)
print(f"{'theta':>6} {'B_par':>6} {'maxRe':>9} {'Gmax':>10} {'t_peak':>7} {'Kreiss':>9} {'PhiF':>10}")
rows=[]
for th in [0,15,30,35,40,45,50,60,75,90]:
    Bp = vA*np.sin(np.radians(th))
    L = Lop(k, Bp)
    Lt = F @ L @ Finv
    mre = eigvals(Lt).real.max()
    Gm, tp = 0.0, 0.0
    if mre < 5e-3:
        for t in ts:
            g = np.linalg.norm(expm(Lt*t), 2)**2
            if g > Gm: Gm, tp = g, t
    else:
        Gm, tp = float('nan'), float('nan')
    Kr = max(z*np.linalg.norm(np.linalg.inv(z*np.eye(2*N)-Lt), 2) for z in zs) if mre < 5e-3 else float('nan')
    Phi = np.linalg.norm(Lt @ Lt.conj().T - Lt.conj().T @ Lt, 'fro')
    rows.append((th,Bp,mre,Gm,tp,Kr,Phi))
print(f"{th:6d} {Bp:6.3f} {mre:9.4f} {Gm:10.1f} {tp:7.1f} {Kr:9.2f} {Phi:10.2f}")

print()
print("=== P5: fastest-growing wavelength at theta=0 (spacing/thickness ratio) ===")
kfine = np.linspace(0.25, 0.7, 19)
sf = [eigvals(Lop(kk,0.0)).real.max() for kk in kfine]
km = kfine[int(np.argmax(sf))]
lam = 2*np.pi/km
print(f" k_max = {km:.3f} -> spacing lambda = {lam:.1f} d (d = tanh half-width)")
print(f" DKIST inversion: spacing 60-100 km -> d = {60/lam:.1f}-{100/lam:.1f} km -> full 90% shear width 4-240 km
d = {240/lam:.0f}-{400/lam:.0f} km... per-unit: {4*60/lam:.0f}-{4*100/lam:.0f} km")

```

## A.2 BH-DYNAMO-01

```

# SCRIPT: BH-DYNAMO-01 (Parker alpha-Omega dynamo operator: oscillatory onset, hair ladder, EP birth of the wave,
# ledger through the EP, Parker-Yoshimura control)
import numpy as np
from scipy.linalg import expm, eig, eigvals
N = 120; L = np.pi
x = np.linspace(0, L, N+2)[1:-1]; h = x[1]-x[0]

```

```

D2 = (np.diag(np.ones(N-1),1) + np.diag(np.ones(N-1),-1) - 2*np.eye(N))/h**2
D1 = (np.diag(np.ones(N-1),1) - np.diag(np.ones(N-1),-1))/(2*h)
alpha_prof = np.cos(x)

def Lop(a0, G=1.0):
    return np.block([[D2, a0*np.diag(alpha_prof)], [G*D1, D2]])

print("=== PD1: onset scan (max Re s vs alpha0; oscillatory?) ===")
a_grid = np.linspace(0.5, 12, 24)
prev = None; a_c = None
for a0 in a_grid:
    w = eigvals(Lop(a0))
    i = np.argmax(w.real)
    if prev is not None and prev < 0 <= w.real[i]:
        a_c = a0
        prev = w.real[i]
if a_c is None: a_c = a_grid[-1]
# refine by bisection
lo, hi = a_c-0.5, a_c
for _ in range(40):
    mid = (lo+hi)/2
    if eigvals(Lop(mid)).real.max() < 0: lo = mid
    else: hi = mid
a_c = hi
w = eigvals(Lop(a_c)); i = np.argmax(w.real)
print(f" alpha0_c = {a_c:.4f} (D_c = a0*G*L^3 = {a_c*np.pi**3:.1f}); leading s = {w[i]:.4f} -> Im = {w[i].imag:.4f} ({'OSCILLATORY onset' if abs(w[i].imag)>1e-6 else 'steady onset'})")
period = 2*np.pi/abs(w[i].imag)

print()
print("=== PD2: the hair ladder at 0.9*alpha0_c (all modes damped) ===")
a0 = 0.9*a_c
Ld = Lop(a0)
w = eigvals(Ld); iw = np.argmax(w.real)
slow = w.real[iw]; cyc = 2*np.pi/abs(w[iw].imag) if abs(w[iw].imag)>1e-9 else np.inf
omega_num = np.linalg.eigvalsh((Ld+Ld.T)/2).max()
Gm, tp = 0.0, 0.0
for t in [0.02,0.05,0.1,0.2,0.35,0.6,1.0,1.6,2.5,4.0]:
    g = np.linalg.norm(expm(Ld*t),2)**2
    if g > Gm: Gm, tp = g, t
Phi = np.linalg.norm(Ld@Ld.T - Ld.T@Ld,'fro')
print(f" SLOW (modal): max Re s = {slow:+.4f}, cycle period = {cyc:.2f}")
print(f" FAST (transient): G_max = {Gm:.1f} at t = {tp} (t_peak/period = {tp/cyc:.3f})")
print(f" HYPER (num. abscissa): omega(L) = {omega_num:+.2f} ({'positive: instantaneous growth in damped ← operator' if omega_num>0 else 'negative'})")
print(f" Phi-stress ||[L,L^T]||_F = {Phi:.1f}")

print()
print("=== PD3: EP hunt -- birth of the wave (real pair -> complex pair) ===")
def top_im(a0):
    w = eigvals(Lop(a0))
    idx = np.argsort(-w.real)[:4]
    return max(abs(w[idx].imag))
lo, hi = 0.05, a_c
assert top_im(lo) < 1e-9 and top_im(hi) > 1e-6
for _ in range(45):
    mid = (lo+hi)/2
    if top_im(mid) < 1e-9: lo = mid
    else: hi = mid
a_ep = (lo+hi)/2
print(f" EP located: alpha0* = {a_ep:.6f} (= {a_ep/a_c:.3f} alpha0_c)")
def kappa_pair(a0):
    Lm = Lop(a0)
    w, vl, vr = eig(Lm, left=True, right=True)
    idx = np.argsort(-w.real)[:2]
    ks = []
    for j in idx:
        num = np.linalg.norm(vl[:,j])*np.linalg.norm(vr[:,j])

```

```

        den = abs(v1[:,j].conj() @ vr[:,j])
        ks.append(num/den)
    return max(ks), w[idx]
print(f" {'delta':>9} {'kappa':>12} {'kappa*sqrt(delta)':>18}")
for d in [0.2, 0.05, 0.0125, 0.003125]:
    kap,_ = kappa_pair(a_ep - d)
    print(f" {d:9.5f} {kap:12.1f} {kap*np.sqrt(d):18.3f}")

print()
print("=== PD4: shifted Dunford ledger through the EP ===")
c = 40.0
for tag, a0 in [("at EP", a_ep), ("near EP", a_ep-0.003)]:
    Lm = Lop(a0)
    w = eigvals(Lm); idx = np.argsort(-w.real)[:2]
    pair = w[idx]
    ctr = pair.real.mean()
    others = np.delete(w, idx)
    r = 0.5*min(abs(others - ctr))
    r = max(r, 2*abs(pair[0]-pair[1]))
    M = 600; th = np.linspace(0, 2*np.pi, M, endpoint=False)
    S = 0.0+0j
    for t in th:
        z = ctr + r*np.exp(1j*t)
        S += np.log(c+z)*np.trace(np.linalg.inv(z*np.eye(2*N)-Lm))*(1j*r*np.exp(1j*t))
    S *= (2*np.pi/M)/(2j*np.pi)
    direct = np.sum(np.log(c+pair))
    print(f" {tag}: ledger = {S.real:.10f} direct sum log(c+pair) = {direct.real:.10f} diff = {abs(S-↔
    direct):.2e}")

print()
print("=== PD5: Parker-Yoshimura control (phase-migration flips with sign of G) ===")
for G in [+1.0, -1.0]:
    Lm = Lop(1.05*a_c, G)
    w, vr = np.linalg.eig(Lm)
    j = np.argmax(w.real)
    B = vr[N:,j]
    ph = np.unwrap(np.angle(B[N//4:3*N//4]))
    slope = np.polyfit(x[N//4:3*N//4], ph, 1)[0]
    print(f" G = {G:+.0f}: leading s = {w[j]:.3f}, mean phase slope d(arg B)/dx = {slope:+.3f}")

```

### A.3 BH-DYNAMO-01b (fix pass)

```

# SCRIPT: BH-DYNAMO-01b (fix pass: true onset, energy norm ladder, exact-pair ledger with count gate, ↔
# Parker-Yoshimura on the wave branch)
import numpy as np
from scipy.linalg import expm, eig, eigvals, cholesky
N = 120; Lx = np.pi
x = np.linspace(0, Lx, N+2)[1:-1]; h = x[1]-x[0]
D2 = (np.diag(np.ones(N-1),1) + np.diag(np.ones(N-1),-1) - 2*np.eye(N))/h**2
D1 = (np.diag(np.ones(N-1),1) - np.diag(np.ones(N-1),-1))/(2*h)
ap = np.cos(x); I = np.eye(N)
def Lop(a0, G=1.0):
    return np.block([[D2, a0*np.diag(ap)], [G*D1, D2]])

print("=== PD1 (fix): true onset, alpha0 up to 80 ===")
def maxRe(a0):
    return eigvals(Lop(a0)).real.max()
a_lo, a_hi = 1.0, 80.0
grid = np.linspace(a_lo, a_hi, 40)
cross = None; prev = maxRe(grid[0])
for a0 in grid[1:]:
    cur = maxRe(a0)
    if prev < 0 <= cur: cross = a0; break
    prev = cur
assert cross is not None, "no onset below 80"
lo, hi = cross - (grid[1]-grid[0]), cross
for _ in range(45):
    mid = (lo+hi)/2

```

```

        if maxRe(mid) < 0: lo = mid
        else: hi = mid
a_c = hi
w = eigvals(Lop(a_c)); j = np.argmax(w.real)
print(f"  alpha0_c = {a_c:.4f}  (D_c = {a_c*np.pi**3:.0f}); leading s = {w[j].real:+.5f} {w[j].imag:+.4f}i ←
      -> {'OSCILLATORY' if abs(w[j].imag)>1e-6 else 'STEADY'} onset; period = {2*np.pi/abs(w[j].imag) if abs(w[j].imag)>1e-9 else float('inf'):.2f}")

print()
print("=== PD2 (fix): hair ladder at 0.9*alpha0_c in the energy norm E = int(|A_x|^2 + |B|^2) ===")
a0 = 0.9*a_c
Ld = Lop(a0)
P = h*(-D2); Wt = np.block([[P, 0*I],[0*I, h*I]])
F = cholesky(Wt, lower=False); Fi = np.linalg.inv(F)
Lt = F @ Ld @ Fi
wl = eigvals(Lt); jj = np.argmax(wl.real)
slow = wl.real[jj]; per = 2*np.pi/abs(wl[jj].imag) if abs(wl[jj].imag)>1e-9 else np.inf
om = np.linalg.eigvalsh((Lt+Lt.conj().T)/2).max()
Gm, tp = 0.0, 0.0
for t in [0.01,0.02,0.04,0.08,0.15,0.3,0.5,0.9,1.5,2.5,4.0,6.0]:
    g = np.linalg.norm(expm(Lt*t),2)**2
    if g > Gm: Gm, tp = g, t
Phi = np.linalg.norm(Lt@Lt.conj().T - Lt.conj().T@Lt, 'fro')/np.linalg.norm(Lt, 'fro')**2
print(f"  SLOW: max Re s = {slow:+.4f}, wave period = {per:.2f}")
print(f"  FAST: G_max = {Gm:.1f} at t = {tp}  (t_peak/period = {tp/per if np.isfinite(per) else 0:.3f})")
print(f"  HYPER: numerical abscissa = {om:+.2f}  (vs slow {slow:+.4f})")
print(f"  normalized Phi-stress = {Phi:.4f}")

print()
print("=== PD4 (fix): ledger with exact-pair contour + enclosed-count gate ===")
a_ep = 3.871725
c = 40.0
for tag, aa in [("at EP", a_ep), ("near EP", a_ep-0.003)]:
    Lm = Lop(aa)
    wv = eigvals(Lm)
    # the colliding pair = two eigenvalues closest to each other among the 6 least-damped
    idx6 = np.argsort(-wv.real)[:6]
    best = None
    for ii in range(6):
        for kk in range(ii+1,6):
            dd = abs(wv[idx6[ii]]-wv[idx6[kk]])
            if best is None or dd < best[0]: best = (dd, idx6[ii], idx6[kk])
    pair = np.array([wv[best[1]], wv[best[2]]])
    ctr = pair.mean()
    others = np.delete(wv, [best[1], best[2]])
    gap = min(abs(others - ctr))
    r = 0.45*gap
    M = 800; th = np.linspace(0,2*np.pi,M,endpoint=False)
    Scount = 0.0+0j; Sled = 0.0+0j
    Iden = np.eye(2*N)
    for t in th:
        z = ctr + r*np.exp(1j*t)
        TrR = np.trace(np.linalg.inv(z*Iden - Lm))
        Scount += TrR*(1j*r*np.exp(1j*t))
        Sled += np.log(c+z)*TrR*(1j*r*np.exp(1j*t))
    Scount *= (2*np.pi/M)/(2j*np.pi); Sled *= (2*np.pi/M)/(2j*np.pi)
    direct = np.sum(np.log(c+pair))
    print(f"  {tag}: enclosed count = {Scount.real:.6f}; ledger = {Sled.real:.10f}; direct = {direct.real:.10f}; diff = {abs(Sled-direct):.2e}")

print()
print("=== PD5 (fix): Parker-Yoshimura on the WAVE branch (least-damped complex mode), G=+-1 ===")
for G in [+1.0, -1.0]:
    Lm = Lop(0.95*a_c, G)
    wv, vr = np.linalg.eig(Lm)
    cand = [j for j in range(len(wv)) if wv[j].imag > 1e-6]
    j = cand[int(np.argmax(wv.real[cand]))]
    B = vr[N:, j]

```

```

seg = slice(N//4, 3*N//4)
ph = np.unwrap(np.angle(B[seg]))
slope = np.polyfit(x[seg], ph, 1)[0]
print(f" G = {G:+.0f}: wave s = {wv[j]:.4f}, phase slope d(arg B)/dx = {slope:+.3f}")

```

## A.4 FIG-SUNS-01 (figures 1–3)

```

# SCRIPT: FIG-SUNS-01 (all figures for the suns paper, regenerated from SHEAR-LANE-01 and BH-DYNAMO-01b)
import numpy as np
import matplotlib
matplotlib.use("Agg")
import matplotlib.pyplot as plt
from scipy.linalg import expm, eig, eigvals, cholesky

# ----- shear-lane machinery -----
N = 200; Ly = 12.0
y = np.linspace(-Ly, Ly, N+2)[1:-1]; h = y[1]-y[0]
U = np.tanh(y); Upp = -2*np.tanh(y)/np.cosh(y)**2
D2 = (np.diag(np.ones(N-1),1)+np.diag(np.ones(N-1),-1)-2*np.eye(N))/h**2
I = np.eye(N); nu = eta = 2e-3; vA = 1.5; k = 0.45
K2 = D2 - k**2*I; K2inv = np.linalg.inv(K2)
P = h*((k**2)*I - D2)
F = cholesky(np.block([[P,0*I],[0*I,P]]), lower=False); Fi = np.linalg.inv(F)
def Llane(Bp):
    L11 = K2inv@(-1j*k*(np.diag(U)@K2) + 1j*k*np.diag(Upp)) + nu*K2
    return np.block([[L11, 1j*k*Bp*I],[1j*k*Bp*I, -1j*k*np.diag(U)+eta*K2]])
ths_all = [0,15,30,35,40,45,50,60,75,90]
Gm, Kr, Ph, MR = [], [], [], []
zs = np.logspace(-2.5,0.3,10); ts = [3,6,10,15,22,32,46,65,90]
for th in ths_all:
    Lt = F @ Llane(vA*np.sin(np.radians(th))) @ Fi
    mre = eigvals(Lt).real.max(); MR.append(mre)
    Ph.append(np.linalg.norm(Lt@Lt.conj().T - Lt.conj().T@Lt,'fro'))
    if mre < 5e-3:
        Gm.append(max(np.linalg.norm(expm(Lt*t),2)**2 for t in ts))
        Kr.append(max(z*np.linalg.norm(np.linalg.inv(z*np.eye(2*N)-Lt),2) for z in zs))
    else:
        Gm.append(np.nan); Kr.append(np.nan)
fig, ax = plt.subplots(1,2, figsize=(9,3.5))
stab = [i for i,t in enumerate(ths_all) if not np.isnan(Gm[i])]
ax[0].plot([ths_all[i] for i in stab],[Gm[i] for i in stab],'bo-',label=r'$G_{\max}$ (energy)')
ax2 = ax[0].twinx()
ax2.plot([ths_all[i] for i in stab],[Kr[i] for i in stab],'rs--',label='Kreiss')
ax[0].axvspan(0,33,color='orange',alpha=0.15)
ax[0].text(6,7.5,'modally\nunstable',fontSize=8)
ax[0].set_xlabel(r'field tilt $\theta$ (deg from vertical)'); ax[0].set_ylabel(r'$G_{\max}$')
ax2.set_ylabel('Kreiss constant')
ax[0].set_title('Directional rigidity: amplification vs tilt')
ax[0].legend(loc='upper right',fontSize=8); ax2.legend(loc='center right',fontSize=8)
ax[1].plot(ths_all, Ph, 'k-')
ax[1].set_xlabel(r'$\theta$ (deg)'); ax[1].set_ylabel(r'$\tilde{L}$\n\tilde{L}\daggers\|_F$')
ax[1].set_title('Stress stored while release falls')
plt.tight_layout(); plt.savefig('figS1_lane.pdf'); plt.close()
print('figS1 done', [f'{g:.1f}' for g in Gm], [f'{p:.2f}' for p in Ph])

# ----- dynamo machinery -----
Nd = 120; Lxx = np.pi
xd = np.linspace(0,Lxx,Nd+2)[1:-1]; hd = xd[1]-xd[0]
D2d = (np.diag(np.ones(Nd-1),1)+np.diag(np.ones(Nd-1),-1)-2*np.eye(Nd))/hd**2
D1d = (np.diag(np.ones(Nd-1),1)-np.diag(np.ones(Nd-1),-1))/(2*hd)
apd = np.cos(xd); Id = np.eye(Nd)
def Ldyn(a0): return np.block([[D2d, a0*np.diag(apd)],[D1d, D2d]])
a_ep = 3.871725
avals = np.linspace(3.4, 4.35, 60)
re1, re2, im1 = [], [], []
for a0 in avals:
    w = eigvals(Ldyn(a0))
    idx6 = np.argsort(-w.real)[:6]

```

```

best=None
for ii in range(6):
    for kk in range(ii+1,6):
        dd=abs(w[idx6[ii]]-w[idx6[kk]])
        if best is None or dd<best[0]: best=(dd,idx6[ii],idx6[kk])
    p=np.array([w[best[1]],w[best[2]]]); p=p[np.argsort(p.real)]
    re1.append(p[0].real); re2.append(p[1].real); im1.append(abs(p[0].imag))
fig, ax = plt.subplots(1,2, figsize=(9,3.5))
ax[0].plot(aval, re1, 'b-', label='Re branch 1'); ax[0].plot(aval, re2, 'b--', label='Re branch 2')
ax[0].plot(aval, im1, 'r-', label='|Im| (wave born)')
ax[0].axvline(a_ep, color='gray', ls=':')
ax[0].set_xlabel(r'$\alpha_0$'); ax[0].set_title(r'EP birth of the dynamo wave at $\alpha_0^*=3.872$')
ax[0].legend(fontsize=8)
ds = [0.2,0.05,0.0125,0.003125]; kaps=[]
for d in ds:
    Lm = Ldyn(a_ep-d)
    w, vl, vr = eig(Lm, left=True, right=True)
    idx = np.argsort(-w.real)[:2]
    kk=[]
    for j in idx:
        kk.append(np.linalg.norm(vl[:,j])*np.linalg.norm(vr[:,j])/abs(vl[:,j].conj()*vr[:,j]))
    kaps.append(max(kk))
ax[1].loglog(ds, kaps, 'ko-', label=r'$\kappa$ measured')
ax[1].loglog(ds, 1.853/np.sqrt(np.array(ds)), 'r--', label=r'$1.853/\sqrt{\delta}$')
ax[1].set_xlabel(r'$\delta = \alpha_0^* - \alpha_0$'); ax[1].set_title(r'Square-root law: $\kappa/\sqrt{\delta} \leftarrow$
    $=1.853$')
ax[1].legend(fontsize=8)
plt.tight_layout(); plt.savefig('figS2_ep.pdf'); plt.close()
print('figS2 done', [f'{q:.1f}' for q in kaps])

# ----- ladder figure -----
a_c = 12.6961; a0 = 0.9*a_c
Pd = hd*(-D2d); Wt = np.block([[Pd,0*Id],[0*Id,hd*Id]])
Fd = cholesky(Wt, lower=False); Fdi = np.linalg.inv(Fd)
Lt = Fd @ Ldyn(a0) @ Fdi
slow = eigvals(Lt).real.max()
om = np.linalg.eigvalsh((Lt+Lt.conj().T)/2).max()
tgrid = np.linspace(0.01, 6, 40)
Gt = [np.linalg.norm(expm(Lt*t),2)**2 for t in tgrid]
fig, ax = plt.subplots(figsize=(5.6,3.6))
ax.semilogy(tgrid, Gt, 'b-', label=r'$G(t)$ (energy norm)')
ax.semilogy(tgrid, np.exp(2*slow*tgrid), 'g--', label=f'slow: modal $2s={2*slow:.2f}$')
ax.semilogy(tgrid[tgrid<0.5], np.exp(2*om*tgrid[tgrid<0.5]), 'r:', label=f'hyper: $2\omega={2*om:.1f}$')
ax.set_xlabel('t'); ax.set_ylabel('energy growth')
ax.set_title(r'The hair ladder at $0.9\alpha_0$: three computed rates')
ax.legend(fontsize=8); plt.tight_layout(); plt.savefig('figS3_ladder.pdf'); plt.close()
print('figS3 done: slow', f'{slow:.4f}', 'omega', f'{om:.2f}', 'Gmax', f'{max(Gt):.1f}')

```

## A.5 N-RECOVERY-01b (the SUN-N-01 loop) and FIG-SUNS-02

```

# SCRIPT: N-RECOVERY-01b (fixes: SF off-grid; EP loop tracking the actual colliding pair, tight radius, ←
    clean contour)
import numpy as np

print("=== G3 (fix): circle spectral flow, crossing off-grid ===")
ks = np.arange(-3,4)
agrid = np.linspace(0.013, 1.013, 2000)
signs_prev = np.sign(ks + agrid[0]); cross = 0
for a in agrid[1:]:
    s = np.sign(ks + a)
    cross += int(np.sum((signs_prev < 0) & (s > 0))) - int(np.sum((signs_prev > 0) & (s < 0)))
    signs_prev = s
print(f"    SF over one period = {cross}")

print()
print("=== G5 (fix): encircling the dynamo EP with the COLLIDING pair, r = 0.15 ===")
N = 120; Lx = np.pi
x = np.linspace(0, Lx, N+2)[1:-1]; h = x[1]-x[0]

```

```

D2 = (np.diag(np.ones(N-1),1)+np.diag(np.ones(N-1),-1)-2*np.eye(N))/h**2
D1 = (np.diag(np.ones(N-1),1)-np.diag(np.ones(N-1),-1))/(2*h)
ap = np.cos(x)
def Ldyn(a0): return np.block([[D2, a0*np.diag(ap)], [D1, D2]])
a_ep = 3.871725; r = 0.15; M = 200
phis = np.linspace(0, 4*np.pi, 2*M+1)
w0 = np.linalg.eigvals(Ldyn(a_ep + r))
idx6 = np.argsort(-w0.real)[:6]
best = None
for i in range(6):
    for j in range(i+1,6):
        d = abs(w0[idx6[i]]-w0[idx6[j]])
        if best is None or d < best[0]: best = (d, idx6[i], idx6[j])
pair = np.array([w0[best[1]], w0[best[2]]]); start = pair.copy()
disc = [np.angle((pair[0]-pair[1])**2)]
after_one = None
for i, ph in enumerate(phis[1:], 1):
    w = np.linalg.eigvals(Ldyn(a_ep + r*np.exp(1j*ph)))
    new = []; used = set()
    for lam in pair:
        j = min((jj for jj in range(len(w)) if jj not in used), key=lambda jj: abs(w[jj]-lam))
        used.add(j); new.append(w[j])
    pair = np.array(new)
    disc.append(np.angle((pair[0]-pair[1])**2))
    if i == M: after_one = pair.copy()
wind = np.unwrap(disc)
swapped = abs(start[0]-after_one[1]) < 1e-3 and abs(start[1]-after_one[0]) < 1e-3
restored = abs(start[0]-pair[0]) < 1e-3 and abs(start[1]-pair[1]) < 1e-3
print(f"    start pair:      {start[0]:.4f}, {start[1]:.4f}")
print(f"    after ONE loop: {after_one[0]:.4f}, {after_one[1]:.4f}    swapped = {swapped}")
print(f"    after TWO loops: {pair[0]:.4f}, {pair[1]:.4f}    restored = {restored}")
print(f"    discriminant winding: one loop = {(wind[M]-wind[0])/(2*np.pi):+.4f}, two loops = {(wind[-1]-wind[0])/(2*np.pi):+.4f}")
c = 40.0
ctr = start.mean(); R = 4*abs(start[0]-start[1]) + 0.6
def contour_pair(a0c):
    Lm = Ldyn(a0c); th = np.linspace(0,2*np.pi,500,endpoint=False)
    Cn = 0j; Ld = 0j; Iden = np.eye(2*N)
    for t in th:
        z = ctr + R*np.exp(1j*t)
        TrR = np.trace(np.linalg.inv(z*Iden - Lm))
        Cn += TrR*(1j*R*np.exp(1j*t)); Ld += np.log(c+z)*TrR*(1j*R*np.exp(1j*t))
    f = (2*np.pi/500)/(2j*np.pi)
    return (Cn*f).real, (Ld*f).real
n0, led0 = contour_pair(a_ep + r)
nh, ledh = contour_pair(a_ep + r*np.exp(1j*np.pi))
n1, led1 = contour_pair(a_ep + r*np.exp(2j*np.pi))
print(f"    contour count at phi=0, pi, 2pi: {n0:.4f}, {nh:.4f}, {n1:.4f}")
print(f"    ledger at phi=0 vs 2pi: {led0:.8f} vs {led1:.8f}    single-valued = {abs(led0-led1)<1e-7}")

# SCRIPT: FIG-SUNS-02 (figS4: the braid at the dynamo EP -- pair exchange and discriminant winding)
import numpy as np, matplotlib
matplotlib.use("Agg")
import matplotlib.pyplot as plt
N=120; Lx=np.pi
x=np.linspace(0,Lx,N+2)[1:-1]; h=x[1]-x[0]
D2=(np.diag(np.ones(N-1),1)+np.diag(np.ones(N-1),-1)-2*np.eye(N))/h**2
D1=(np.diag(np.ones(N-1),1)-np.diag(np.ones(N-1),-1))/(2*h)
ap=np.cos(x)
Ldyn=lambda a0: np.block([[D2,a0*np.diag(ap)], [D1,D2]])
a_ep=3.871725; r=0.15; M=200
phis=np.linspace(0,4*np.pi,2*M+1)
w0=np.linalg.eigvals(Ldyn(a_ep+r)); idx6=np.argsort(-w0.real)[:6]
best=None
for i in range(6):
    for j in range(i+1,6):
        d=abs(w0[idx6[i]]-w0[idx6[j]])
        if best is None or d<best[0]: best=(d,idx6[i],idx6[j])

```

```

pair=np.array([w0[best[1]],w0[best[2]]]); start=pair.copy()
tr1=[pair[0]]; tr2=[pair[1]]; disc=np.angle((pair[0]-pair[1])**2)
for ph in phis[1:]:
    w=np.linalg.eigvals(Ldyn(a_ep+r*np.exp(1j*ph)))
    new=[];used=set()
    for lam in pair:
        j=min((jj for jj in range(len(w)) if jj not in used),key=lambda jj:abs(w[jj]-lam))
        used.add(j); new.append(w[jj])
    pair=np.array(new); tr1.append(pair[0]); tr2.append(pair[1])
    disc.append(np.angle((pair[0]-pair[1])**2))
tr1=np.array(tr1); tr2=np.array(tr2); wind=np.unwrap(disc)
fig,ax=plt.subplots(1,2,figsize=(9,3.6))
ax[0].plot(tr1[:M+1].real,tr1[:M+1].imag,'b-',lw=1.5,label='branch 1, loop 1')
ax[0].plot(tr2[:M+1].real,tr2[:M+1].imag,'r-',lw=1.5,label='branch 2, loop 1')
ax[0].plot(start[0].real,start[0].imag,'bo',ms=8)
ax[0].plot(start[1].real,start[1].imag,'rs',ms=8)
ax[0].plot(tr1[M].real,tr1[M].imag,'b*',ms=13)
ax[0].plot(tr2[M].real,tr2[M].imag,'r*',ms=13)
ax[0].set_xlabel(r'Re $\lambda$'); ax[0].set_ylabel(r'Im $\lambda$')
ax[0].set_title('One loop around the EP: the pair exchanges')
ax[0].legend(fontsize=8)
ax[1].plot(phis/(2*np.pi),(wind-wind[0])/(2*np.pi),'k-')
ax[1].axhline(1,color='gray',ls=':'); ax[1].axhline(2,color='gray',ls=':')
ax[1].set_xlabel('loops around the EP'); ax[1].set_ylabel(r'winding of $\arg(\lambda_1-\lambda_2)^{2/2\pi}$')
ax[1].set_title('Integer record: winding $+1$ per loop')
plt.tight_layout(); plt.savefig('figS4_braid.pdf'); plt.close()
print("figS4 done: winding one/two loops =", (wind[M]-wind[0])/(2*np.pi), (wind[-1]-wind[0])/(2*np.pi))

```

## A.6 SUN-OPEN-01 (closures) and FIG-SUNS-03

```

# SCRIPT: SUN-OPEN-01 (closing open problems 1,2,3,8: hemispheric Parker-Yoshimura, onset character, k↔
# optimized envelope, EP/braid census)
import numpy as np
from scipy.linalg import expm, eigvals, cholesky

# ----- hemispheric Parker model: x in [0, pi/2], pole Dirichlet, equator dipole parity -----
N = 120; Lx = np.pi/2
x = np.linspace(0, Lx, N+1)[1:] # interior + last node adjacent to equator ghost
h = x[1]-x[0]
def ops_dirichlet():
    D2 = (np.diag(np.ones(N-1),1)+np.diag(np.ones(N-1),-1)-2*np.eye(N))/h**2
    D1 = (np.diag(np.ones(N-1),1)-np.diag(np.ones(N-1),-1))/(2*h)
    return D2, D1
# B: Dirichlet pole and equator. A: Dirichlet pole, Neumann equator (dipole parity)
D2B, D1B = ops_dirichlet()
D2A = D2B.copy(); D1A = D1B.copy()
D2A[-1,-1] = -1/h**2 # ghost A_{N+1}=A_N
D1A[-1,-1] = 1/(2*h) # (A_N - A_{N-1})/2h at last row
ap = np.cos(x)
def Lhemi(a0, G=1.0):
    return np.block([[D2A, a0*np.diag(ap)], [G*D1A, D2B]])

print("=== OP2: onset character of the hemispheric (dipole-parity) model ===")
def maxRe(a0, G=1.0):
    return eigvals(Lhemi(a0,G)).real.max()
grid = np.linspace(0.5, 60, 60); cross=None; prev=maxRe(grid[0])
for a0 in grid[1:]:
    cur = maxRe(a0)
    if prev < 0 <= cur: cross=a0; break
    prev=cur
lo, hi = cross-1.0, cross
for _ in range(40):
    mid=(lo+hi)/2
    if maxRe(mid)<0: lo=mid
    else: hi=mid
a_c = hi
w = eigvals(Lhemi(a_c)); j=np.argmax(w.real)
osc = abs(w[j].imag) > 1e-6

```

```

print(f"    alpha0_c(hemi) = {a_c:.4f}; leading s = {w[j].real:+.5f} {w[j].imag:+.4f}i -> {'OSCILLATORY' if ←
    osc else 'STEADY'} onset"
    + (f", cycle period {2*np.pi/abs(w[j].imag):.2f}" if osc else ""))

print()
print("=== OP1: Parker-Yoshimura flip in the hemispheric model ===")
for G in [+1.0, -1.0]:
    a0 = 0.98*a_c
    wv, vr = np.linalg.eig(Lhemi(a0, G))
    cand = [k for k in range(len(wv)) if wv[k].imag > 1e-6]
    j = cand[int(np.argmax(wv.real[cand]))]
    B = vr[N:, j]
    seg = slice(N//5, 4*N//5)
    ph = np.unwrap(np.angle(B[seg]))
    slope = np.polyfit(x[seg], ph, 1)[0]
    print(f"    G = {G:+.0f}: wave s = {wv[j]:.4f}, phase slope d(arg B)/dx = {slope:+.3f}")

# butterfly data for the figure (G=+1 wave mode)
a0 = 0.98*a_c
wv, vr = np.linalg.eig(Lhemi(a0, +1.0))
cand = [k for k in range(len(wv)) if wv[k].imag > 1e-6]
j = cand[int(np.argmax(wv.real[cand]))]
np.save('butterfly_mode.npy', np.array([wv[j].real, wv[j].imag]))
np.save('butterfly_B.npy', vr[N:, j]); np.save('butterfly_x.npy', x)

print()
print("=== OP8: EP/braid census along alpha0 (full-domain model) ===")
Nd=120; Lxx=np.pi
xd=np.linspace(0,Lxx,Nd+2)[1:-1]; hd=xd[1]-xd[0]
D2d=(np.diag(np.ones(Nd-1),1)+np.diag(np.ones(Nd-1),-1)-2*np.eye(Nd))/hd**2
D1d=(np.diag(np.ones(Nd-1),1)-np.diag(np.ones(Nd-1),-1))/(2*hd)
apd=np.cos(xd)
Ldyn=lambda a0: np.block([[D2d, a0*np.diag(apd)],[D1d, D2d]])
def n_complex_top(a0, k=10):
    w = eigvals(Ldyn(a0)); idx = np.argsort(-w.real)[:k]
    return int(np.sum(w[idx].imag > 1e-7))
eps_list=[]
agrid = np.linspace(0.2, 13.0, 129)
prev = n_complex_top(agrid[0])
for a0 in agrid[1:]:
    cur = n_complex_top(a0)
    if cur > prev:
        lo, hi = a0-0.1, a0
        for _ in range(35):
            mid=(lo+hi)/2
            if n_complex_top(mid) <= prev: lo=mid
            else: hi=mid
        eps_list.append((hi, cur-prev))
    prev = max(prev, cur)
print(f"    pair-birth events found: {[round(a,4) for a,_ in eps_list]}")
def winding_at(aep, r=0.10, M=80):
    w0 = eigvals(Ldyn(aep + r)); idx6 = np.argsort(-w0.real)[:8]
    best=None
    for i in range(8):
        for jj in range(i+1,8):
            d=abs(w0[idx6[i]]-w0[idx6[jj]])
            if best is None or d<best[0]: best=(d,idx6[i],idx6[jj])
    pair=np.array([w0[best[1]],w0[best[2]]]); start=pair.copy()
    disc=[np.angle((pair[0]-pair[1])**2)]
    for ph in np.linspace(0,2*np.pi,M+1)[1:]:
        w=eigvals(Ldyn(aep + r*np.exp(1j*ph)))
        new=[];used=set()
        for lam in pair:
            k2=min((kk for kk in range(len(w)) if kk not in used),key=lambda kk:abs(w[kk]-lam))
            used.add(k2); new.append(w[k2])
        pair=np.array(new); disc.append(np.angle((pair[0]-pair[1])**2))
    wind=np.unwrap(disc)
    swapped = abs(start[0]-pair[1])<5e-3 and abs(start[1]-pair[0])<5e-3

```

```

        return (wind[-1]-wind[0])/(2*np.pi), swapped
print("  census (alpha0*, winding per loop, pair exchanged):")
for aep,_ in eps_list[:4]:
    wnd, sw = winding_at(aep)
    print(f"      alpha0* = {aep:8.4f}    winding = {wnd:+.3f}    exchange = {sw}")

print()
print("=== OP3 (2D part): k-optimized transient envelope of the shear lane ===")
Ns=200; Ly=12.0
y=np.linspace(-Ly,Ly,Ns+2)[1:-1]; hs=y[1]-y[0]
U=np.tanh(y); Upp=-2*np.tanh(y)/np.cosh(y)**2
D2s=(np.diag(np.ones(Ns-1),1)+np.diag(np.ones(Ns-1),-1)-2*np.eye(Ns))/hs**2
Is=np.eye(Ns); nu=eta=2e-3; vA=1.5
def Gmax_lane(th_deg, k):
    K2=D2s-k**2*Is; K2i=np.linalg.inv(K2)
    Bp=vA*np.sin(np.radians(th_deg))
    L11=K2i@(-1j*k*(np.diag(U)*K2)+1j*k*np.diag(Upp))+nu*K2
    L=np.block([[L11,1j*k*Bp*Is],[1j*k*Bp*Is,-1j*k*np.diag(U)+eta*K2]])
    P=hs*((k**2)*Is-D2s)
    F=cholesky(np.block([[P,0*Is],[0*Is,P]]),lower=False); Fi=np.linalg.inv(F)
    Lt=F@L@Fi
    if eigvals(Lt).real.max() > 5e-3: return np.nan
    return max(np.linalg.norm(expm(Lt*t),2)**2 for t in [5,10,18,30,50,80])
for th in [35, 45, 60]:
    best=(0,0)
    for k in [0.15,0.25,0.35,0.45,0.6,0.8]:
        g=Gmax_lane(th,k)
        if not np.isnan(g) and g>best[0]: best=(g,k)
    print(f"      theta={th}: G_max over k = {best[0]:.1f} at k={best[1]}    (fixed-k value at 0.45 was {Gmax_lane(th,0.45):.1f})")

print()
print("=== OP5 (theory side): spacing/width ratio vs tilt in the unstable band ===")
for th in [0, 10, 20, 30]:
    Bp=vA*np.sin(np.radians(th))
    ks=np.linspace(0.2,0.8,13); sig=[]
    for k in ks:
        K2=D2s-k**2*Is; K2i=np.linalg.inv(K2)
        L11=K2i@(-1j*k*(np.diag(U)*K2)+1j*k*np.diag(Upp))+nu*K2
        L=np.block([[L11,1j*k*Bp*Is],[1j*k*Bp*Is,-1j*k*np.diag(U)+eta*K2]])
        sig.append(eigvals(L).real.max())
    km=ks[int(np.argmax(sig))]
    print(f"      theta={th}: k_max={km:.3f} -> spacing = {2*np.pi/km:.1f} half-widths (spacing/full-width = ←
        {2*np.pi/km/4:.2f})")

# SCRIPT: FIG-SUNS-03 (figS5: hemispheric butterfly diagrams -- Parker-Yoshimura flip)
import numpy as np, matplotlib
matplotlib.use("Agg")
import matplotlib.pyplot as plt
N=120; Lx=np.pi/2
x=np.linspace(0,Lx,N+1)[1:]; h=x[1]-x[0]
D2B=(np.diag(np.ones(N-1),1)+np.diag(np.ones(N-1),-1)-2*np.eye(N))/h**2
D1B=(np.diag(np.ones(N-1),1)-np.diag(np.ones(N-1),-1))/(2*h)
D2A=D2B.copy(); D1A=D1B.copy()
D2A[-1,-1]=-1/h**2; D1A[-1,-1]=1/(2*h)
ap=np.cos(x)
Lh=lambda a0,G: np.block([[D2A,a0*np.diag(ap)],[G*D1A,D2B]])
a0=0.98*12.5264
fig,ax=plt.subplots(1,2,figsize=(9,3.6))
for col,G in enumerate([+1.0,-1.0]):
    ww,vr=np.linalg.eig(Lh(a0,G))
    cand=[k for k in range(len(ww)) if ww[k].imag>1e-6]
    j=cand[int(np.argmax(ww.real[cand]))]
    B=vr[N:,j]; om=ww[j].imag
    T=2*np.pi/om
    t=np.linspace(0,2*T,240)
    lat=90-x*180/np.pi
    F=np.real(np.outer(np.exp(1j*om*t),B))

```

```

F=F/np.max(np.abs(F))
ax[col].contourf(t/T,lat,F.T,levels=21,cmap='RdBu_r')
ax[col].set_xlabel('t / cycle'); ax[col].set_ylabel('latitude (deg)')
ax[col].set_title(f'G={G:+.0f}: '+'(poleward' if G>0 else 'equatorward (solar-like)')')
plt.tight_layout(); plt.savefig('figS5_butterfly.pdf'); plt.close()
print("figS5 done")

```

## A.7 SUN-OPEN-02 and SUN-OPEN-02b (full census, shell onset)

```

# SCRIPT: SUN-OPEN-02 (v1.3 audit: EP4/5 with radial-continuation tracking; 2D shell onset mechanism; ←
# hazard-law arithmetic)
import numpy as np
from scipy.linalg import eigvals

print("=== A: EP census with radial-continuation pair selection (all five points) ===")
N=120; Lx=np.pi
x=np.linspace(0,Lx,N+2)[1:-1]; h=x[1]-x[0]
D2=(np.diag(np.ones(N-1),1)+np.diag(np.ones(N-1),-1)-2*np.eye(N))/h**2
D1=(np.diag(np.ones(N-1),1)-np.diag(np.ones(N-1),-1))/(2*h)
ap=np.cos(x)
Ldyn=lambda a0: np.block([[D2,a0*np.diag(ap)],[D1,D2]])
def newborn_pair(a0, ktop=14):
    w=eigvals(Ldyn(a0)); idx=np.argsort(-w.real)[:ktop]
    cpx=[j for j in idx if w[j].imag>1e-9]
    j=min(cpx,key=lambda j: w[j].imag) # smallest positive-Im = just-born branch
    lam=w[j]
    jj=min((k for k in idx if k!=j),key=lambda k:abs(w[k]-np.conj(lam)))
    return np.array([lam,w[jj]])
def track(pair, a0c):
    w=eigvals(Ldyn(a0c)); new=[];used=set()
    for lam in pair:
        k=min((kk for kk in range(len(w)) if kk not in used),key=lambda kk:abs(w[kk]-lam))
        used.add(k); new.append(w[k])
    return np.array(new)
for a_ep in [3.8717,5.2802,7.1576,9.0850,11.0320]:
    r=0.12
    pair=newborn_pair(a_ep+0.004)
    for aa in np.linspace(a_ep+0.004,a_ep+r,18)[1:]: # radial continuation out to loop radius
        pair=track(pair,aa)
    start=pair.copy(); disc=[np.angle((pair[0]-pair[1])**2)]
    for ph in np.linspace(0,2*np.pi,101)[1:]:
        pair=track(pair,a_ep+r*np.exp(1j*ph))
        disc.append(np.angle((pair[0]-pair[1])**2))
    wind=(np.unwrap(disc)[-1]-np.unwrap(disc)[0])/(2*np.pi)
    sw = abs(start[0]-pair[1])<5e-3 and abs(start[1]-pair[0])<5e-3
    print(f" alpha0* = {a_ep:8.4f}: winding = {wind:+.4f} exchange = {sw}")

print()
print("=== B: 2D shell onset mechanism (radial structure) ===")
def shell_onset(Nr, Nx, shear): # shear: 'lat' = G dA/dx (control), 'rad' = G dA/dr (solar alpha-Omega ←
# geometry)
    rr=np.linspace(0.7,1.0,Nr+2)[1:-1]; hr=rr[1]-rr[0]
    xx=np.linspace(0,np.pi,Nx+2)[1:-1]; hx=xx[1]-xx[0]
    D2r=(np.diag(np.ones(Nr-1),1)+np.diag(np.ones(Nr-1),-1)-2*np.eye(Nr))/hr**2
    D2x=(np.diag(np.ones(Nx-1),1)+np.diag(np.ones(Nx-1),-1)-2*np.eye(Nx))/hx**2
    D1r=(np.diag(np.ones(Nr-1),1)-np.diag(np.ones(Nr-1),-1))/(2*hr)
    D1x=(np.diag(np.ones(Nx-1),1)-np.diag(np.ones(Nx-1),-1))/(2*hx)
    Ir,Ix=np.eye(Nr),np.eye(Nx)
    Lap=np.kron(D2r,Ix)+np.kron(Ir,D2x)
    Alpha=np.kron(Ir,np.diag(np.cos(xx)))
    Sh = np.kron(Ir,D1x) if shear=='lat' else np.kron(D1r,Ix)
    def Lsh(a0): return np.block([[Lap,a0*Alpha],[Sh,Lap]])
    def mR(a0): return eigvals(Lsh(a0)).real.max()
    grid=np.linspace(5,240,24); cross=None; prev=mR(grid[0])
    for a0 in grid[1:]:
        cur=mR(a0)
        if prev<0<=cur: cross=a0; break
    prev=cur

```

```

    if cross is None: return None
    lo,hi=cross-(grid[1]-grid[0]),cross
    for _ in range(28):
        mid=(lo+hi)/2
        if mR(mid)<0: lo=mid
        else: hi=mid
    w=eigvals(Lsh(hi)); j=np.argmax(w.real)
    return hi, w[j]
for shear in ['lat','rad']:
    res = shell_onset(16,32,shear)
    if res is None: print(f"    shear={shear}: no onset below a0=240"); continue
    ac,s = res
    print(f"    shear={shear} (16x32): a0c = {ac:8.3f}, leading s = {s.real:+.5f} {s.imag:+.4f}i -> {'←'
        'OSCILLATORY' if abs(s.imag)>1e-5 else 'STEADY'}")
res = shell_onset(22,44,'rad')
if res:
    ac,s = res
    print(f"    shear=rad (22x44): a0c = {ac:8.3f}, leading s = {s.real:+.5f} {s.imag:+.4f}i -> {'←'
        'OSCILLATORY' if abs(s.imag)>1e-5 else 'STEADY'} (resolution check)")

print()
print("=== C: hazard-law arithmetic and sensitivity ===")
for beta in [-0.8,-0.9,-1.0]:
    if beta== -1.0: print(f"    beta={beta:+.1f}: exponent -> infinity (logarithmic hazard)"); continue
    print(f"    beta={beta:+.1f}: WTD tail exponent 3+beta = {3+beta:.1f}; hazard lambda ~ Phi^(1/(1+beta)) ←
        = Phi^{1/(1+beta):.0f}")
print(f"    steepness at exponent 10: 7.2% stress rise doubles the rate (2**0.1={2**0.1:.4f}); 26% rise -> x←
    {1.26**10:.1f}")
print()
print("=== D: OP5 width-per-tilt table (arithmetic from R(theta)) ===")
for th,Rr in [(0,3.49),(10,3.93),(20,4.49),(30,6.28)]:
    print(f"    theta={th:2d}: width for 60 km spacing = {60/Rr:.1f} km; for 100 km = {100/Rr:.1f} km")

# SCRIPT: SUN-OPEN-02b (shell onset pinned: radial-shear oscillatory onset + latitudinal-shear control + ←
    resolution check)
import numpy as np
from scipy.linalg import eigvals
def build(Nr,Nx,shear,r0=0.2):
    rr=np.linspace(r0,1.0,Nr+2)[1:-1]; hr=rr[1]-rr[0]
    xx=np.linspace(0,np.pi,Nx+2)[1:-1]; hx=xx[1]-xx[0]
    D2r=(np.diag(np.ones(Nr-1),1)+np.diag(np.ones(Nr-1),-1)-2*np.eye(Nr))/hr**2
    D2x=(np.diag(np.ones(Nx-1),1)+np.diag(np.ones(Nx-1),-1)-2*np.eye(Nx))/hx**2
    D1r=(np.diag(np.ones(Nr-1),1)-np.diag(np.ones(Nr-1),-1))/(2*hr)
    D1x=(np.diag(np.ones(Nx-1),1)-np.diag(np.ones(Nx-1),-1))/(2*hx)
    Ir,Ix=np.eye(Nr),np.eye(Nx)
    Lap=np.kron(D2r,Ix)+np.kron(Ir,D2x)
    Alpha=np.kron(Ir,np.diag(np.cos(xx)))
    Sh=np.kron(Ir,D1x) if shear=='lat' else np.kron(D1r,Ix)
    return lambda a0: np.block([[Lap,a0*Alpha],[Sh,Lap]])
def mR(L,a0): return eigvals(L(a0)).real.max()
L=build(16,32,'rad')
lo,hi=1500.,2500.
for _ in range(24):
    mid=(lo+hi)/2
    if mR(L,mid)<0: lo=mid
    else: hi=mid
w=eigvals(L(hi)); j=np.argmax(w.real)
print(f"RADIAL shear (16x32): a0c = {hi:.1f}, onset s = {w[j].real:+.4f}{w[j].imag:+.3f}i -> {'OSCILLATORY' ←
    if abs(w[j].imag)>1e-4 else 'STEADY'}, cycle = {2*np.pi/abs(w[j].imag):.4f}")
L1=build(16,32,'lat')
for a0 in [1000,2500,5000]:
    w=eigvals(L1(a0)); j=np.argmax(w.real)
    print(f"LATITUDINAL shear control: a0={a0}: max Re = {w[j].real:+.3f}, Im = {w[j].imag:+.3f}")
L2=build(22,44,'rad')
w=eigvals(L2(hi)); j=np.argmax(w.real)
print(f"resolution check (22x44) at a0c(16x32): s = {w[j].real:+.3f}{w[j].imag:+.3f}i ({'oscillatory branch←
    persists' if abs(w[j].imag)>1e-3 else 'steady?!'})")

```

## A.8 DKIST-COMPARE-02 (the comparison of §3)

```
#!/usr/bin/env python3
# SCRIPT: DKIST-COMPARE-02
# The comparison run in the convention-free direction: from the independently
# computed shear-layer width to the predicted wavelength.
#
# SHEAR-LANE-01 fixes ONE dimensionless number with no observational input:
#     k_max * d = 0.450    =>    lambda / d = 2*pi/0.450 = 13.96
# d is the tanh half-width in U = tanh(y/d) and is convention-free.
# WHAT WAS AN INPUT: the observed spacings, used in the v1 width inversion.
# WHAT WAS NOT: the growth rates and the companion layer width, below.
# What is NOT convention-free is how a paper reports "shear layer width" from a
# tanh profile. Standard choices span 2d (scale width) to 4d (96% of the jump).
# So their 12 km maps to a BAND of d, hence a BAND of predicted wavelength.
import numpy as np
LAM_D = 2*np.pi/0.450          # 13.963, fixed by the computation
W_MURAM = 12.0                  # km, companion MURaM shear layer
LAM_OBS, LAM_LO, LAM_HI = 65.0, 25.0, 170.0 # peak and full observed span
print("=*76); print("DKIST-COMPARE-02  their width -> our wavelength"); print("=*76)
print(f"\n fixed by the computation alone: lambda/d = {LAM_D:.3f}")
print(f"\n independently computed layer: W_MURaM = {W_MURAM:.0f} km")
print(f"\n observed wavelength:          peak {LAM_OBS:.0f} km (span {LAM_LO:.0f}--{LAM_HI:.0f})")
print(f"\n {'convention':<34} {'d (km)':>8} {'predicted lambda (km)':>22} {'vs 65':>9}")
convs=[("2d  scale width",2.0),("2.94d 90% of the jump",2.94),
        ("3d  common tanh reporting",3.0),("4d  96% of the jump",4.0)]
preds=[]
for name,f in convs:
    d=W_MURAM/f; lam=LAM_D*d; preds.append(lam)
    print(f"  {name:<34} {d:>8.2f} {lam:>22.1f} {lam/LAM_OBS:>8.2f}x")
lo,hi=min(preds),max(preds)
print(f"\n predicted band from their 12 km: {lo:.1f} - {hi:.1f} km")
print(f"\n observed peak 65 km lies inside the band: {lo <= LAM_OBS <= hi}")
print(f"\n the convention that would make it exact: W = {W_MURAM/(LAM_OBS/LAM_D):.2f} d "
      f"(between the 2d and 4d conventions)")

print("\n RUN THE OTHER WAY (our wavelength -> our width), for the record:")
d_from_obs = LAM_OBS/LAM_D
print(f"  observed lambda 65 km => d = {d_from_obs:.2f} km")
print(f"  => 2d = {2*d_from_obs:.1f} km  2.94d = {2.94*d_from_obs:.1f} km  4d = {4*d_from_obs:.1f} km")
print(f"  their 12 km sits at {W_MURAM/d_from_obs:.2f} d, inside that range")

print("\n WHAT IS ACTUALLY FALSIFIABLE, and is not yet tested:")
print("  the ratio lambda/d = 13.96 is a PREDICTION about the mode, not a fit.")
print("  Publishing d under a stated convention turns the band above into a")
print("  single number and the comparison into a pass or a fail.")
print(f"\n growth-rate cross-check (independent of all width conventions):")
sig_obs=(0.014,0.054)          # s^-1, measured
U_lo,U_hi=0.67e3,3.0e3         # m/s apparent speeds
d_m=d_from_obs*1e3
print(f"  Michalke peak for a tanh layer: sigma_max ~ 0.19 * (dU/2) / d")
for U in (U_lo,U_hi):
    print(f"    dU = {U/1e3:.2f} km/s -> sigma = {0.19*(U/2)/d_m:.4f} s^-1")
print(f"  measured range: {sig_obs[0]}--{sig_obs[1]} s^-1")
print("=*76)
```

## Acknowledgments

Computational verification, gating, and drafting assistance by an AI collaborator (Claude, Anthropic); every gate above is independently re-runnable from Appendix A. License: CC BY 4.0.

## References

- [1] D. Kuridze et al., Ubiquitous Kelvin–Helmholtz instabilities driving plasma mixing on the Sun, *Nature* (2026), doi:10.1038/s41586-026-10871-3; MPS/NSF–NSO press release 26900660.
- [2] K. Nykyri, Photospheric Kelvin–Helmholtz vortices as possible drivers of coronal heating: implications of the DKIST observations, arXiv:2608.12796 (2026).
- [3] The striated solar photosphere observed at 0.03'' resolution, *Astrophys. J. Lett.* (2025); arXiv:2505.03965.
- [4] M. J. Aschwanden and T. Dudok de Wit, Waiting-time distributions of solar flares, coronal mass ejections, and radio bursts, *Astrophys. J.* (2021).
- [5] M. S. Wheatland, Understanding solar flare waiting-time distributions, *Solar Phys.* (2003).
- [6] L. Biasiotti and S. L. Ivanovski, Statistical analysis of solar flare properties from 1975 to 2017, *Solar Phys.* **300**, 121 (2025).
- [7] Waiting time distribution of solar flares from a global perspective, *Astrophys. J. Lett.* (2026); arXiv:2604.00877.
- [8] A convolutional neural network–derived catalog of solar flares from soft X-ray observations, *Astrophys. J. Suppl.* (2026).
- [9] T. R. Rimmele et al., The Daniel K. Inouye Solar Telescope: Observatory overview, *Solar Phys.* **295**, 172 (2020).
- [10] A. E. Fraser et al., Nonmodal growth and optimal perturbations in magnetohydrodynamic shear flows, *Phys. Rev. E* (2026); arXiv:2510.03141.
- [11] A. Michalke, On the inviscid instability of the hyperbolic-tangent velocity profile, *J. Fluid Mech.* **19**, 543 (1964).
- [12] S. Chandrasekhar, *Hydrodynamic and Hydromagnetic Stability*, Oxford (1961).
- [13] R. Keppens, G. Tóth, R. H. J. Westermann, and J. P. Goedbloed, Growth and saturation of the Kelvin–Helmholtz instability with parallel and anti-parallel magnetic fields, *J. Plasma Phys.* **61**, 1 (1999).
- [14] M. Obergaulinger et al., Local simulations of the magnetized Kelvin–Helmholtz instability in neutron-star mergers, *Astron. Astrophys.* **515**, A30 (2010).
- [15] D. MacTaggart, Optimal energy growth in current sheets, *Solar Phys.* **293**, 148 (2018).
- [16] L. N. Trefethen, A. E. Trefethen, S. C. Reddy, and T. A. Driscoll, Hydrodynamic stability without eigenvalues, *Science* **261**, 578 (1993).
- [17] L. N. Trefethen and M. Embree, *Spectra and Pseudospectra*, Princeton (2005).
- [18] B. F. Farrell and P. J. Ioannou, Generalized stability theory. Part I, *J. Atmos. Sci.* **53**, 2025 (1996).
- [19] E. N. Parker, Hydromagnetic dynamo models, *Astrophys. J.* **122**, 293 (1955).

- [20] H. Yoshimura, Solar-cycle dynamo wave propagation, *Astrophys. J.* **201**, 740 (1975).
- [21] P. Charbonneau, Dynamo models of the solar cycle, *Living Rev. Sol. Phys.* **17**, 4 (2020).
- [22] L. Chen, W. Herreman, K. Li, P. W. Livermore, J. W. Luo, and A. Jackson, Optimal kinematic dynamos in a sphere, *Proc. R. Soc. A* **476**, 20190675 (2020).
- [23] W. Herreman, Minimal perturbation flows that trigger mean field dynamos in shear flows, *J. Plasma Phys.* **84**, 735840305 (2018).
- [24] O. N. Kirillov, *Nonconservative Stability Problems of Modern Physics*, De Gruyter (2013).
- [25] J. L. Jaramillo, R. Panosso Macedo, and L. Al Sheikh, Pseudospectrum and black hole quasinormal mode instability, *Phys. Rev. X* **11**, 031003 (2021).
- [26] E. Gasperin and J. L. Jaramillo, Energy scales and black hole pseudospectra: the structural role of the scalar product, *Class. Quantum Grav.* **39**, 115010 (2022).
- [27] J. Besson, J. Carballo, C. Pantelidou, and B. Withers, Transients in black hole perturbation theory, *Front. Phys.* **13**, 1638583 (2025).
- [28] J. Carballo, C. Pantelidou, and B. Withers, Transient superradiance, *JHEP* **08**, 179 (2025).
- [29] H. Li and F. D. M. Haldane, Entanglement spectrum as a generalization of entanglement entropy, *Phys. Rev. Lett.* **101**, 010504 (2008).
- [30] H. Hu and E. Zhao, Knots and non-Hermitian Bloch bands, *Phys. Rev. Lett.* **126**, 010401 (2021).
- [31] C. C. Wojcik, X.-Q. Sun, T. Bzdušek, and S. Fan, Homotopy characterization of non-Hermitian Hamiltonians, *Phys. Rev. B* **101**, 205417 (2020).
- [32] C. Dembowski et al., Experimental observation of the topological structure of exceptional points, *Phys. Rev. Lett.* **86**, 787 (2001).
- [33] T. Kato, *Perturbation Theory for Linear Operators*, Springer (1966).
- [34] J. Levine, Knot cobordism groups in codimension two, *Comment. Math. Helv.* **44**, 229 (1969).
- [35] A. G. Tristram, Some cobordism invariants for links, *Proc. Camb. Phil. Soc.* **66**, 251 (1969).
- [36] J. N. Beasley, Foundational Operator Field Theory on Stratified Manifolds, v2 (2026), companion paper.
- [37] J. N. Beasley, Beyond the Event Horizon: Operator Decoupling at the Rigidity Wall, v2 (2026), companion paper.